# Repetitionless

**The offline documentation can be hard to follow so using the online documentation is recommended which you can find at https://docs.wilschack.dev/repetitionless. You can still use this if you need though, it has all the same information**

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# Table of contents


1. Introduction 4
   1.1 Features Video 4
   1.2 Description 4
   1.3 Contents 4
2. Installation 5
   2.1 Download 5
   2.2 Installation 5
3. Getting Started With Repetitionless 6
   3.1 Tutorial Video 6
   3.2 Welcome Screen 6
   3.3 Sample Scenes 7
   3.4 Using A Repetitionless Material 7
   3.5 Painting 8
   3.6 Integrations 8
   3.7 Creating Shaders 8
   3.8 Creating Scripts 8
4. Samples 9
   4.1 Opening The Samples 9
   4.2 Sample Settings 10
5. Material Details 12
   5.1 Data Folder 12
   5.2 Texture Packing 12
   5.3 Colour Space 12
6. Converting Materials 14
   6.1 Converting A Material 14
7. Using Regular Materials 16
   7.1 Creating A Material 16
   7.2 Using The Material 17
   7.3 Configuration 17
8. Using Layered Materials 18
   8.1 Creating A Layered Material 18
   8.2 Using Terrain Layers 19
   8.3 Using Control Textures 24
9. Material Properties 26
   9.1 Material Properties 26




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9.2 Toolbar 27
9.3 Material Settings 28
9.4 Distance Blending 30
9.5 Material Blending 31
9.6 Debug 33
10. Painting 34
10.1 Tutorial Video 34
10.2 Setting Up 34
10.3 Selection 35
10.4 Layers 35
10.5 Holes 36
10.6 Properties 38
10.7 Keybinds 39
10.8 Caveats 40
11. Shader Creation 42
11.1 Shader Graphs 42
11.2 Shader Code 42
12. Scripting 43
13. Submitting Issues 44
14. Integrations 45
14.1 MapMagic 45
15. Sub Graphs 0
15.1 Compression 0
15.2 Distance Blending 0
15.3 Macro Micro Variation 0
15.4 Noise 0
16. Shader API 0
16.1 Main 0
16.2 Noise 0
16.3 RepetitionlessHelpers 0
16.4 Utilities 0
16.5 Structs 0
16.6 TextureArrayEssentials 0
17. Scripting API 0
17.1 Runtime 0
17.2 Editor 0


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# 1. Introduction

## 1.1 Features Video

https://youtu.be/BqwM6js7TkU

## 1.2 Description

Repetitionless is a set of shaders that removes texture tiling using multiple different toggleable techniques including:

- Voronoi noise and cell edge sampling to split up the texture and smooth between cells
- Random scaling and rotation between cells
- Triplanar sampling
- Variation texture support to add variation to the material
- Distance blending to either change the tiling and offset or material entirely at a set distance range
- Material Blending to overlay a separate material ontop of the main one by different noise functions or a custom mask

## 1.3 Contents


- Installation
- Getting Started
- Samples
- Material Details
- Converting Materials
- Using Regular Materials
- Using Layered Materials
- Material Properties
- Shader Creation
- Scripting
- Submitting Issues




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# 2. Installation

## 2.1 Download

**Requirements**:

- Unity 2021.3+

Repetitionless can be downloaded from:

- Unity Asset Store
- Itch.io

## 2.2 Installation

### 2.2.1 Unity Asset Store

Install it the same you would any other unity package:

1. Open the Unity Package Manager
2. Select Repetitionless under My Assets
3. Click Download, then Import after it is finished

### 2.2.2 Itch.io

You can install the package by either:

- Dragging the .unitypackage file into the project window
- Importing it through `Assets > Import Package > Custom Package`



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# 3. Getting Started With Repetitionless

## 3.1 Tutorial Video

https://youtu.be/Ujk6ZyCmWak

## 3.2 Welcome Screen

When first importing the asset you will be greeted with a welcome screen with buttons to:

- View Documentation
- View and Reporting issues
- Import samples

To open this window again, you can find it in the toolbar at
`Window > Repetitionless > Open Window`


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## 3.3 Sample Scenes

The asset comes with 3 example scenes that you can view to see the different features of repetitionless and how to implement the materials

For instructions on accessing the samples:

**View the Samples page**

## 3.4 Using A Repetitionless Material

**There are details you need to know when using the materials:**

**View the Material Details Page**

For instructions on converting regular lit materials to repetitionless materials:

**View the Converting Materials Page**

For instructions on how to use the regular repetitionless shader:



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**View the Using Regular Materials Page**

For instructions on how to use the layered repetitionless shader or a terrain:

**View the Using Layered Materials Page**

To view what each material property does:

**View the Material Properties Page**

## 3.5 Painting

If you want details on how to paint layers onto objects:

**View the Painting Page**

## 3.6 Integrations

**If you want to use any of the asset integrations, details are in their respective pages:**

For instructions on using Repetitionless terrain materials on a MapMagic terrain:

**View the MapMagic Page**

## 3.7 Creating Shaders

You can view the Sub Graphs and Shader API documentation in the contents on the left of the page

If you want to create a shader graph or code shader using the features from the repetitionless materials:

**View the Shader Creation page**

## 3.8 Creating Scripts

You can view the Sub Graphs and Shader API documentation in the contents on the left of the page

If you want to create scripts or inspectors using the features from the repetitionless materials inspectors:

**View the Scripting page**

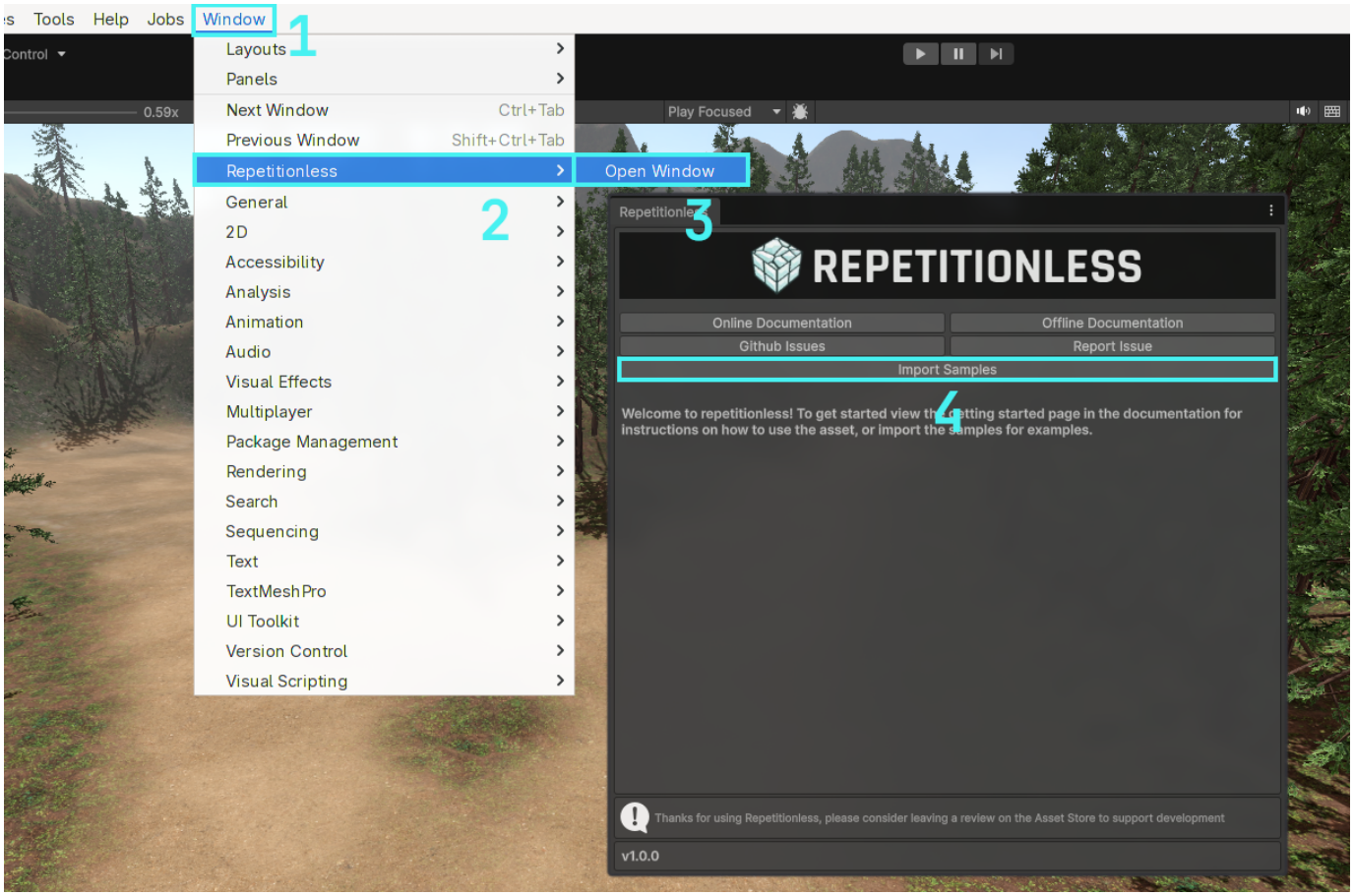


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# 4. Samples

Samples are located in a folder which is hidden by default. To import the samples:

1. Open the windows tab in the toolbar
2. Navigate to `Repetitionless`
3. Click `Open Window`
4. Click the import samples button and it will automatically import your current render pipelines samples

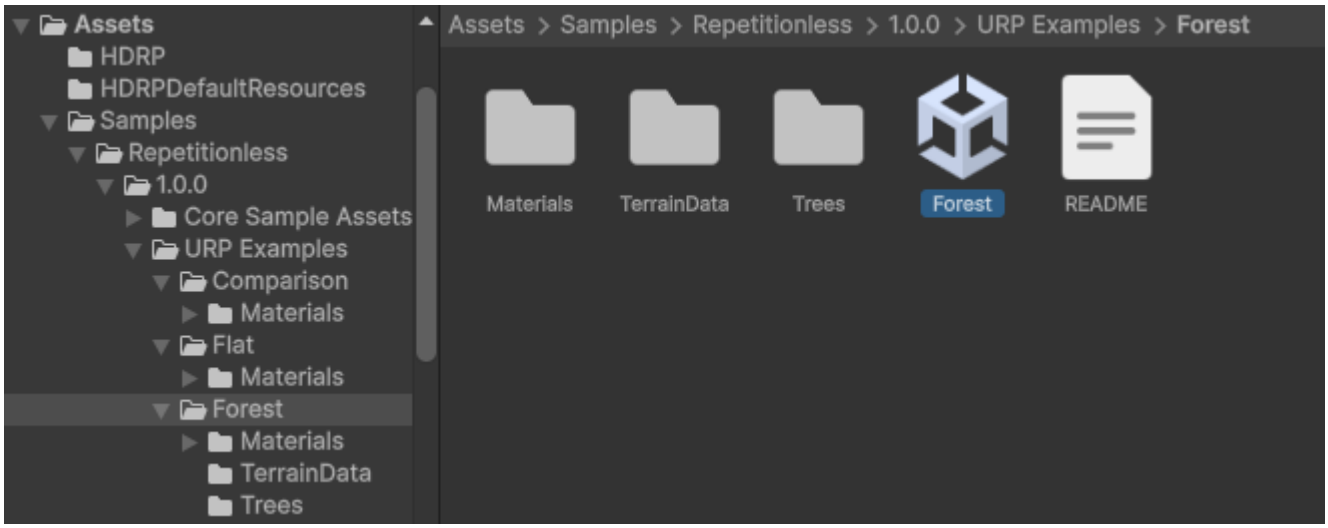



## 4.1 Opening The Samples

Samples will now be located in
`Assets/Samples/Repetitionless/<CurrentVersion>/<RP> Examples`

To open the sample scenes you can go to their respective folders and open the scene


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*The Core Sample Assets folder stores the base assets for all the samples, this must be imported for the samples to work*

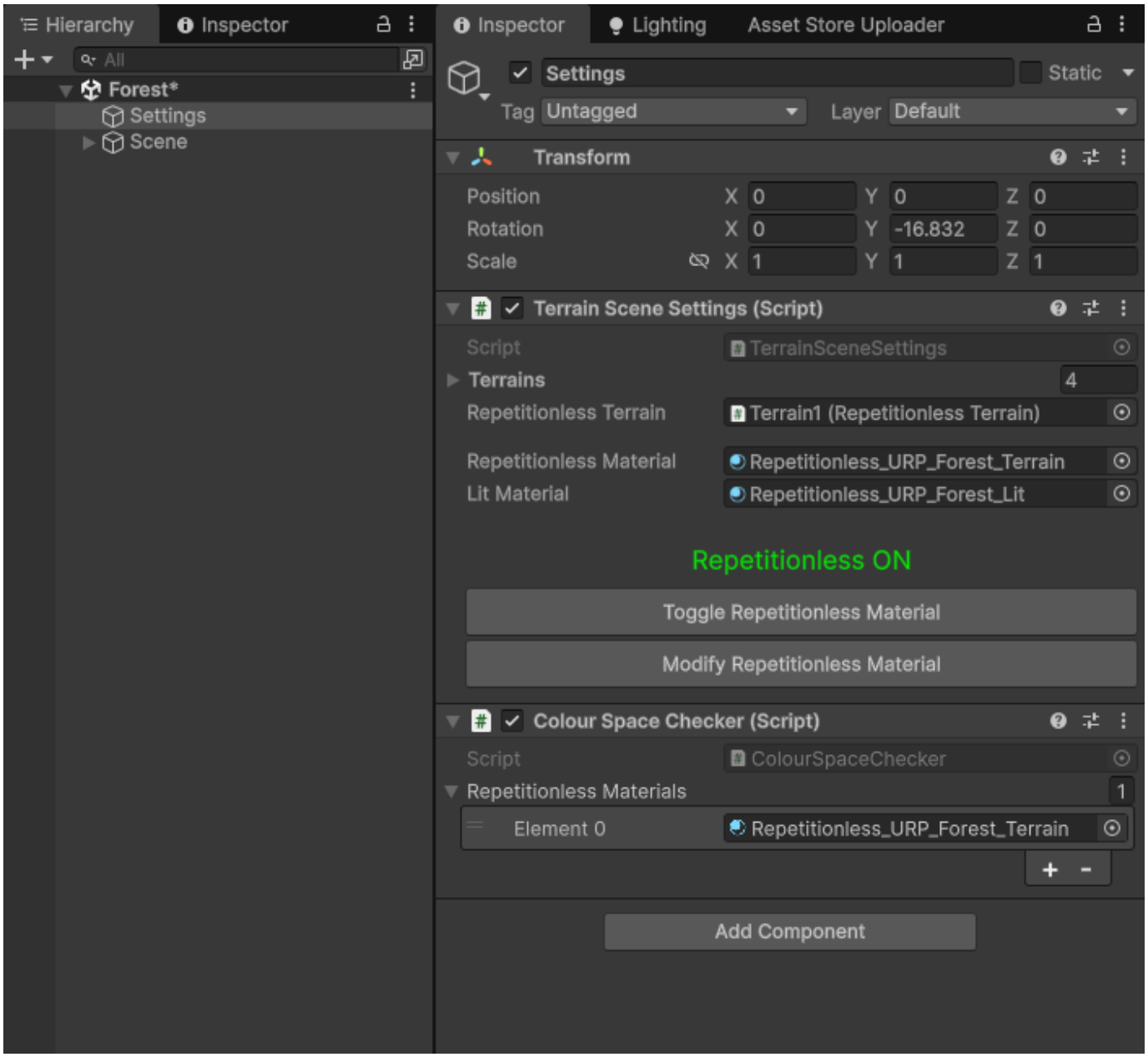
## 4.2 Sample Settings

The samples have a settings object that has buttons to:

- Toggle between the repetitionless and regular lit materials
- Modify and view the material settings used



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*(The Colour Space Checker is used to make sure the textures are packed in the correct colour space, it is not needed after the scene is opened for the first time)*



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# 5. Material Details

## 5.1 Data Folder

All the properties and textures are stored in a folder along side the material named
`<Material Name>_RepetitionlessData`

This folder is automatically handled and you do not need to worry about it

*This folder will automatically be moved and deleted with the material but will not be copied so if you need to copy a material you will need to manually copy the folder aswell*




## 5.2 Texture Packing

All the textures are packed into texture arrays for less samples in the shader which are stored in the data folder along side the material. These can be modified by clicking the **Array Settings button** in the Material Properties section

Arrays are split into:

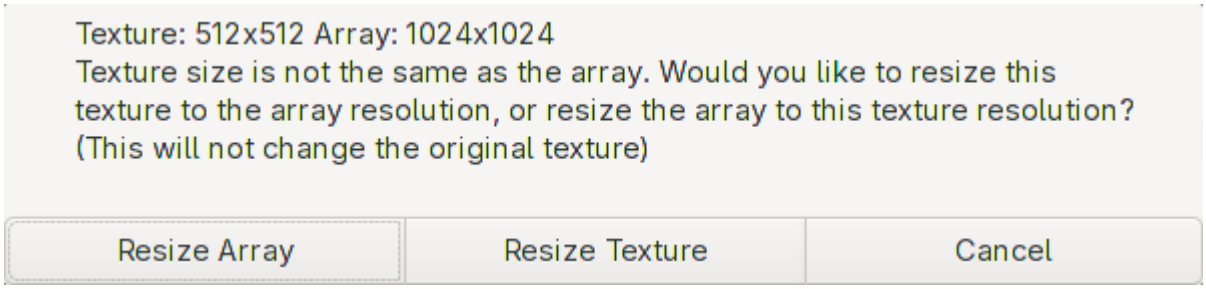
- AVTextures: Albedo (rgb), Variation (a)
- NSOTextures: Normal (rg), Smoothness (b), Occlussion (a)
- EMTextures: Emission (rgb), Metallic (a)
- BMTextures: Blend Mask (r)

*Only the required arrays are created, and only the used textures are stored*

The downside to texture arrays is that they can only have one resolution each array of textures, but this is automatically handled. This is also per array so for example the albedo textures can be a different resolution to the normal textures

When you assign a texture that is not the same resolution as the array, there is a popup that lets you either:

- Resize all the textures in the array to the resolution of the new texture
- Resize the texture to the same resolution as the array

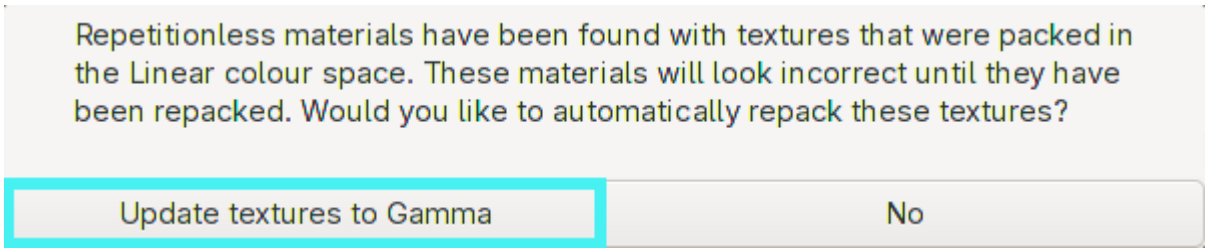



## 5.3 Colour Space

sRGB textures (Albedo/Colour) are packed with the set colour space in mind. They will need to be repacked when changing the colour space

There is a popup when changing the colour space if any textures need to be repacked that will automatically repack the textures for you.
Make sure to select **Update Textures to \<Colour Space>** for the textures to look correct


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***To view what each property does, visit the Material Properties page***



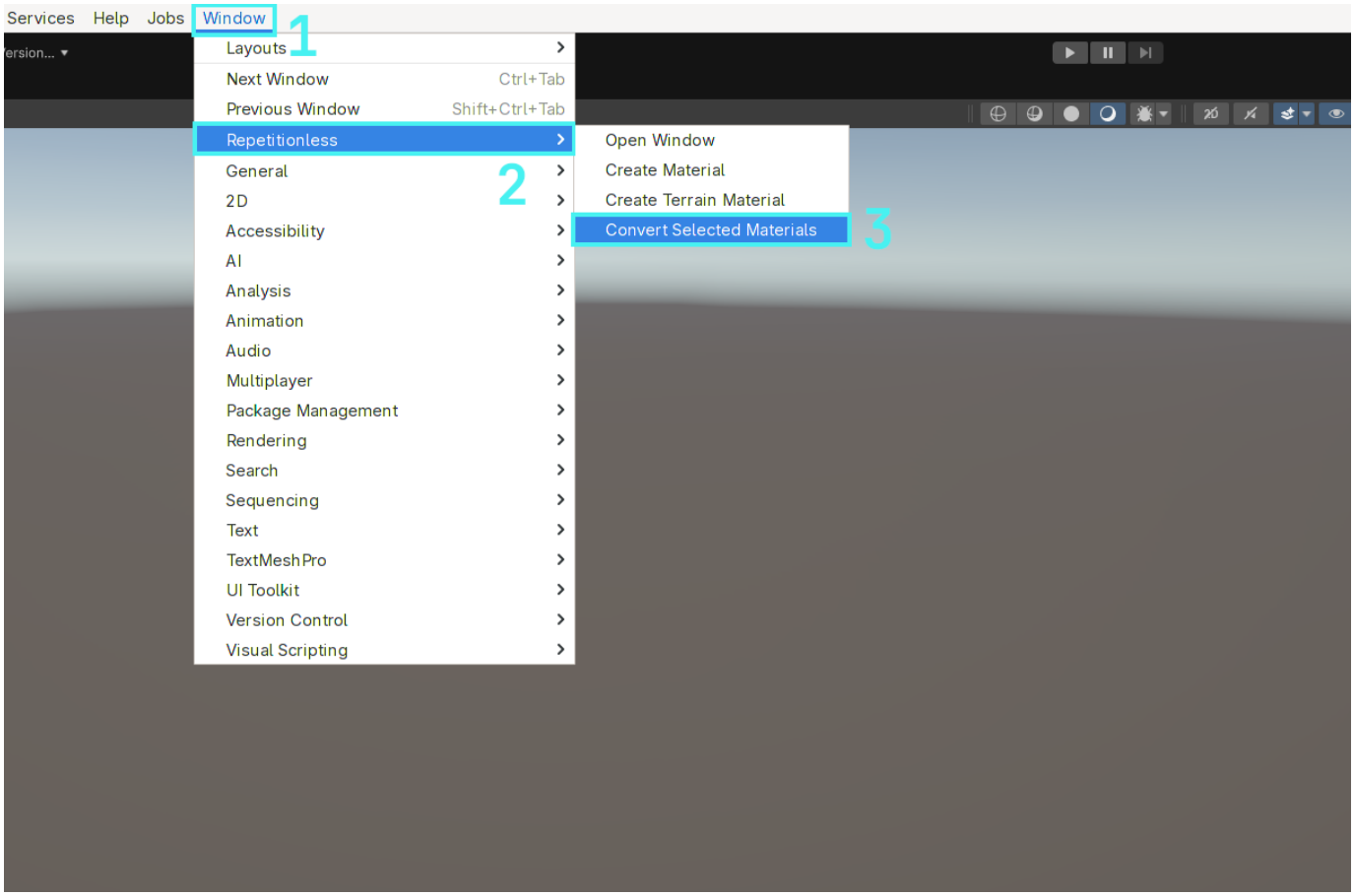
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# 6. Converting Materials

## 6.1 Converting A Material

To convert a regular lit material to a repetitionless material:

1. Select the materials you want to convert
2. Open the `Window` tab in the toolbar, or the `Create` menu in the project window
3. Navigate to `Repetitionless`
4. Click `Convert Selected Materials`
5. The materials will be created next to the original materials with the suffix "_repetitionless"








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**Important Details:**

- This will always convert to the same render pipeline as the input material
- This works for BIRP, URP, & HDRP standard lit materials, any others will not work

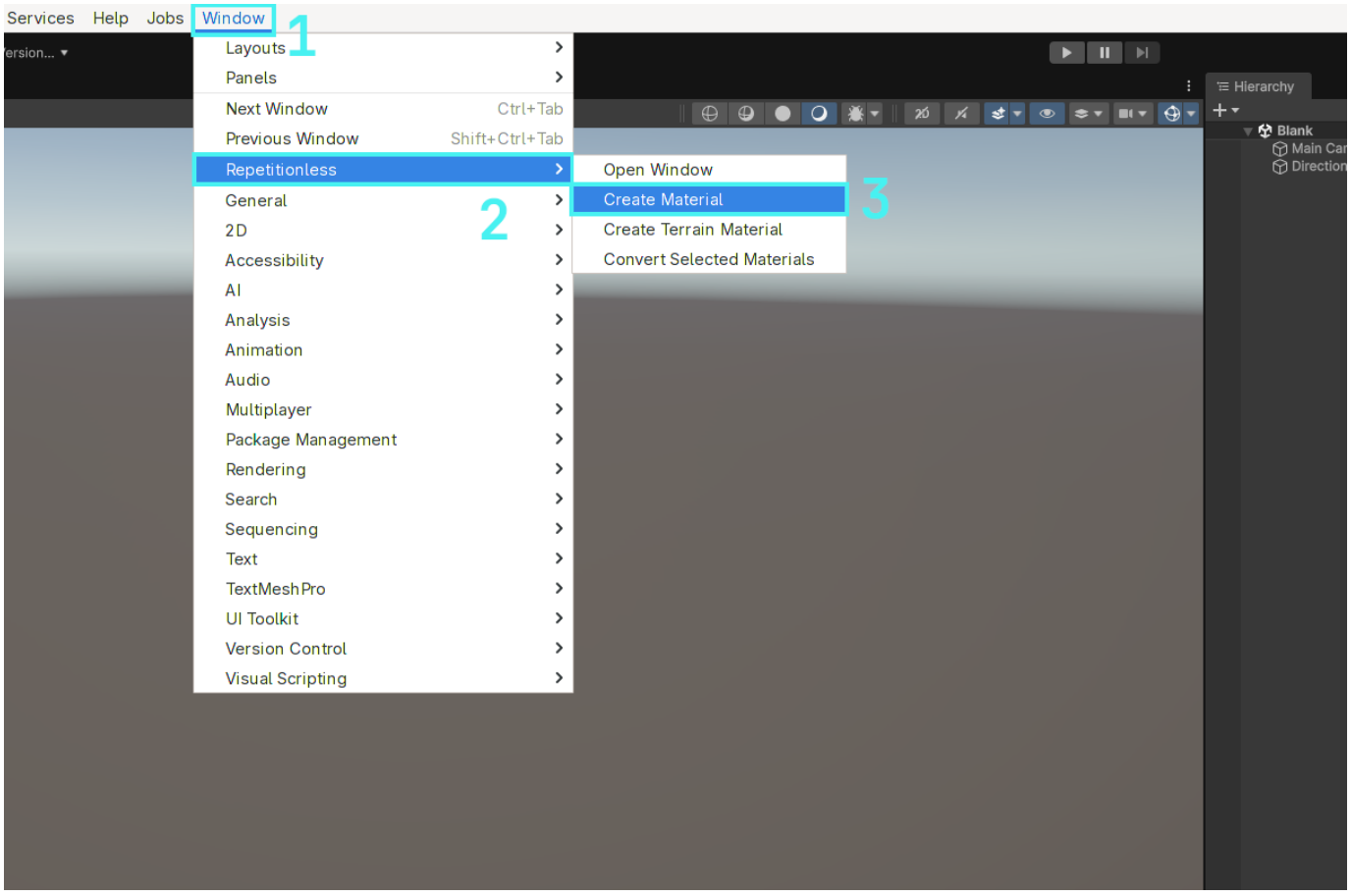

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# 7. Using Regular Materials

## 7.1 Creating A Material

### 7.1.1 Automatic

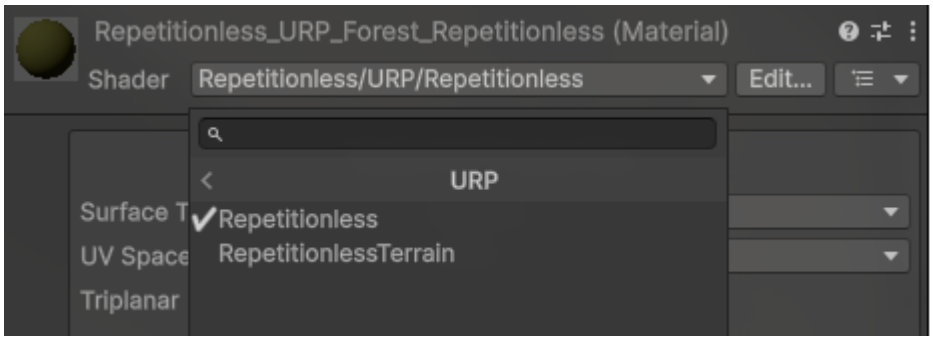
1. Open the `Window` tab in the toolbar, or the `Create` menu in the project window
2. Navigate to `Repetitionless`
3. Click `Create Material`
4. The material will be created in the current folder in the project window




### 7.1.2 Manual

1. Create a material
2. Select the shader dropdown
3. Navigate to `Repetitionless`
4. Select your render pipeline
5. Select `Repetitionless`


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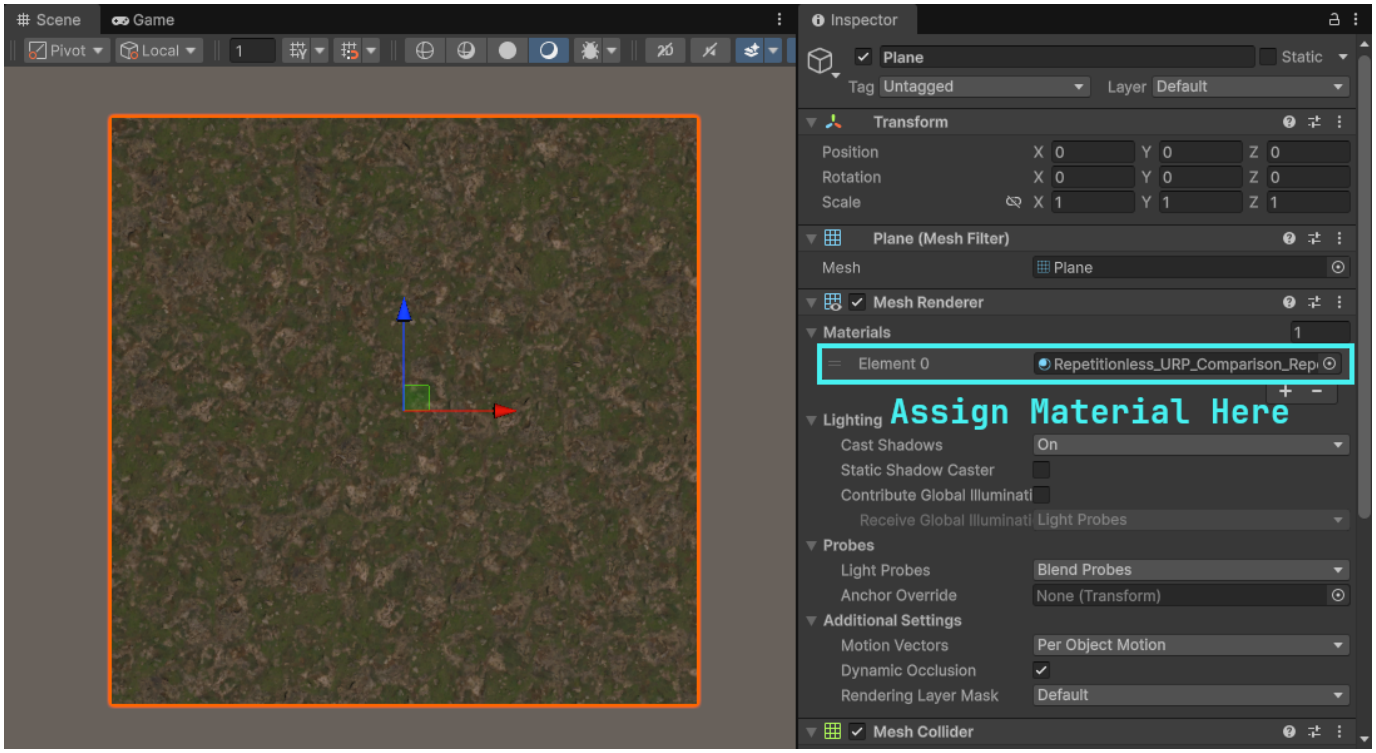



**Important Details:**

- Render pipelines can be switched without losing data. ex. Changing a Repetitionless material from BIRP to its URP version will keep all the settings
- Material data is stored in a folder along side the material named "{MaterialName}_Data". This is automatically moved and deleted with the material but it will need to be manually copied when copying the material

## 7.2 Using The Material

You can just assign repetitionless material to any Mesh Renderer and it will be applied to that object




## 7.3 Configuration

All the materials can be configured through its inspector by selecting the material and viewing the inspector window

***To view what each property does, visit the Material Properties page***



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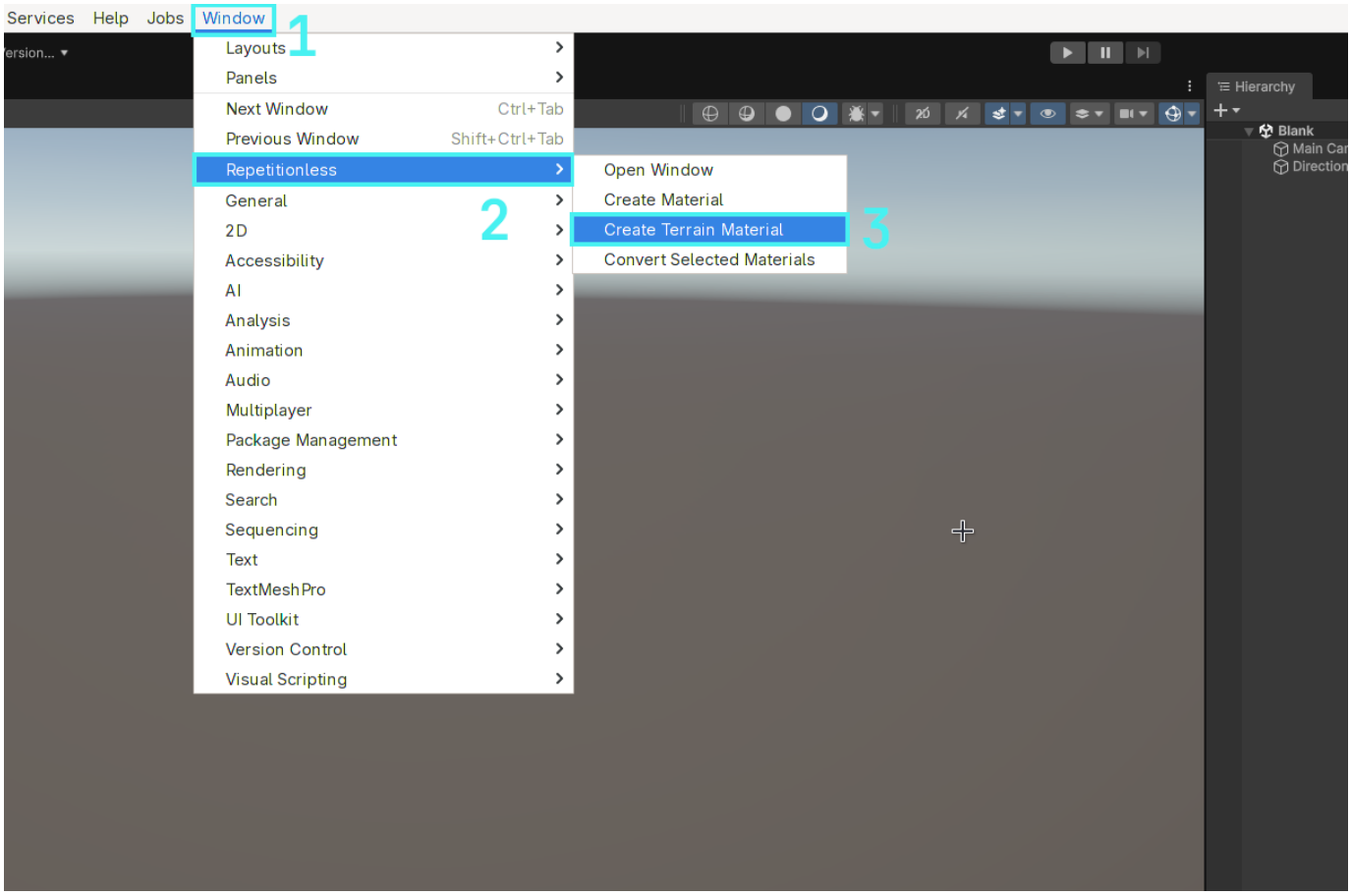
# 8. Using Layered Materials

The layered material is used to display up to 32 different material layers on one material. Layer positions can either be updated with control textures or synced with a terrain. Instructions are below for both

## 8.1 Creating A Layered Material

### 8.1.1 Automatic

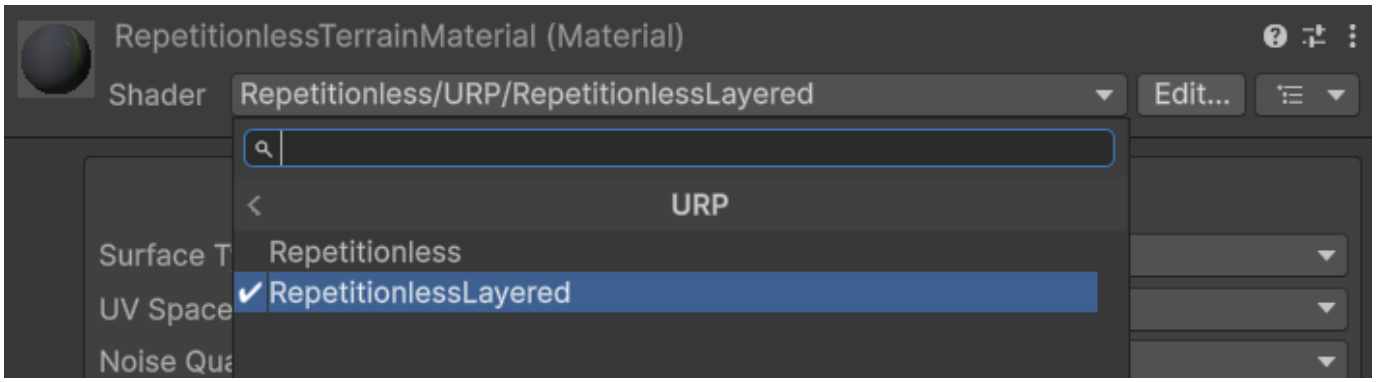
1. Open the `Window` tab in the toolbar, or the `Create` menu in the project window
2. Navigate to `Repetitionless`
3. Click `Create Layered Material`
4. The material will be created in the current folder in the project window




### 8.1.2 Manual

1. Create a material
2. Select the shader dropdown
3. Navigate to `Repetitionless`
4. Select your render pipeline
5. Select `RepetitionlessLayeredTerrain` or `RepetitionlessLayeredLit`


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**Important Details:**

- Render pipelines can be switched without losing data. ex. Changing a Repetitionless material from BIRP to its URP version will keep all the settings
- Material data is stored in a folder along side the material named "{MaterialName}_RepetitionlessData". This is automatically moved and deleted with the material but it will need to be manually copied when copying the material
- The `LayeredTerrain` shader is used when the mode is set to Terrain Layers, `LayeredLit` is used when the mode is set to anything else. This can be changed in the inspector, so when creating the material you can select either.

## 8.2 Using Terrain Layers

To use a terrain with the repetitionless material you have to add a `RepetitionlessTerrain` component to the terrain which can be done as follows:

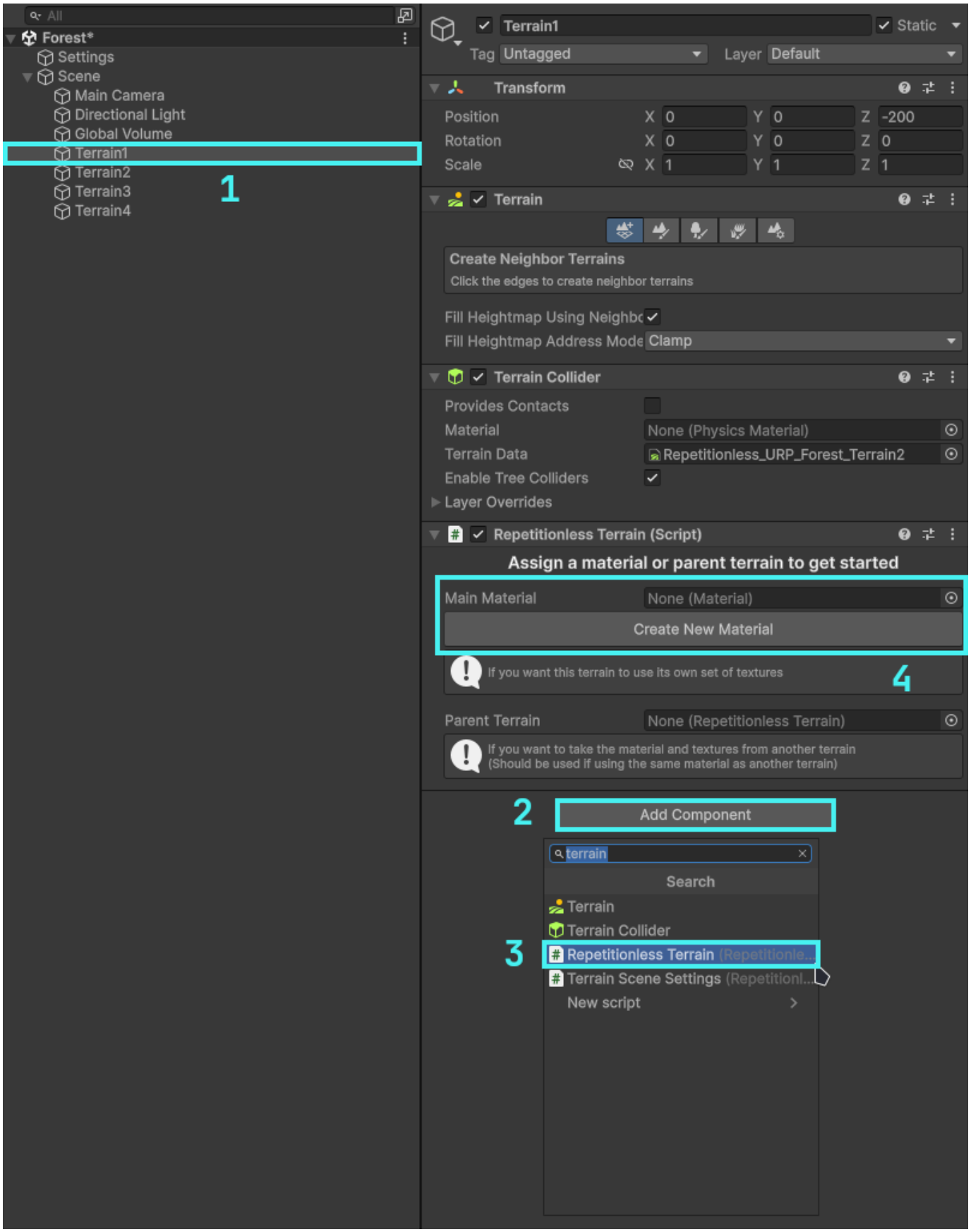
1. Select the terrain you want to use
2. Select `Add Component`
3. Select `Repetitionless Terrain`

To add a material you can either:

- Click the `Create New Material` button
- Assign a material in the `Main Material` field

***Note the the material field will only allow Repetitionless Layered materials. Assigning a material will also automatically set its layer mode to Terrain Layers***

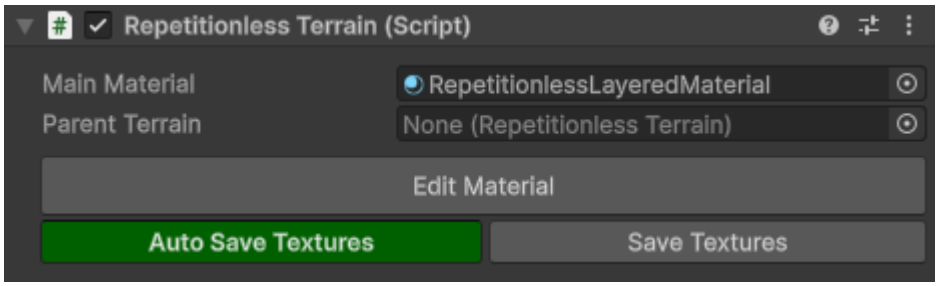

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## 8.2.1 Using The Terrain Script



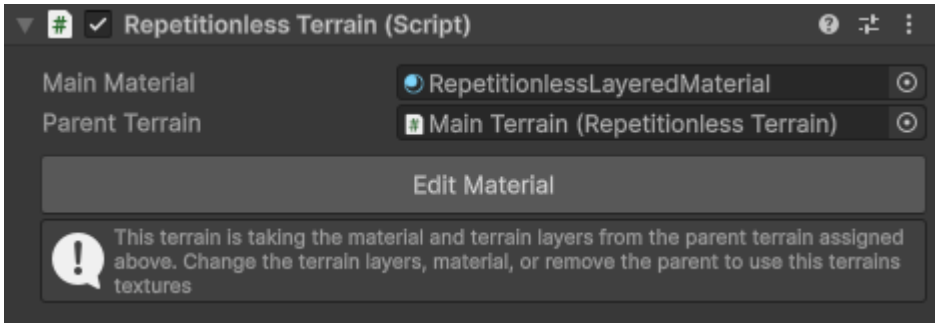

| Setting | Description |
|---|---|
| Main Material | The material applied to the terrains |
| Parent Terrain | The terrain to take the material and terrain layers from. See below for details |
| Edit Material | Opens the inspector for the selected material |
| Auto Save Textures | Toggles if the material layers are automatically updated if the terrain layers change |
| Save Textures | Reapplies the terrain materials if required, and syncs the terrain layers to the material. Click this if you have any issues with the terrain |

That is basically all you need to do! The script will handle the material assigning and texture syncing for you

With Auto Save Textures enabled, the terrain will sync whenever you update the terrain layers, updating the materials textures. With it disabled, it will not sync the textures until you click the Save Textures button.

**If you ever have any errors with the terrain, first see if clicking the Save Textures button fixes your issue. If it doesnt feel free to submit a bug report**

## 8.2.2 Using Multiple Terrains



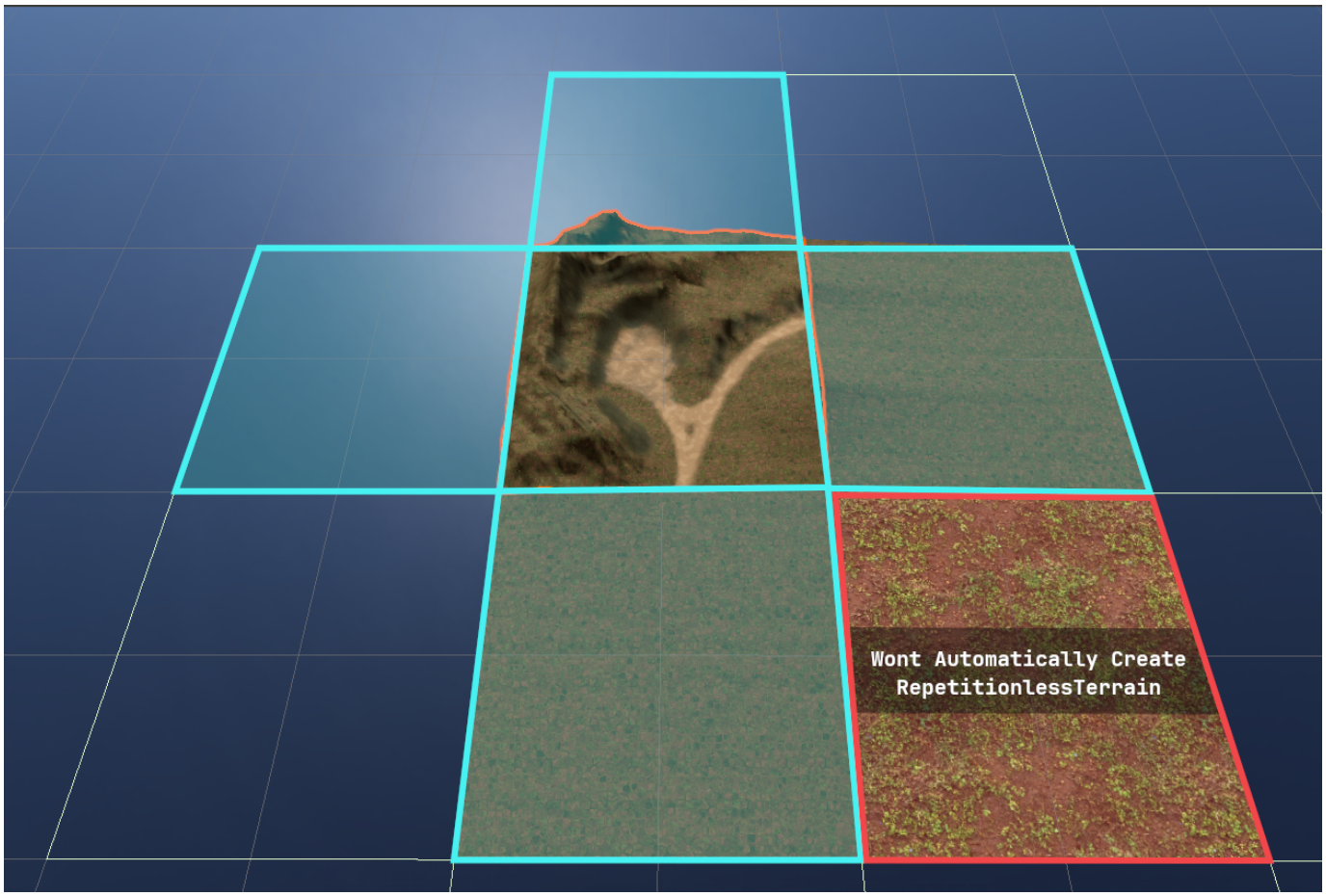

The script also has a Parent Terrain field that when set will update that terrain to use the same material and layers from the parent. Keep in mind that when setting a parent terrain it will overwrite the current terrain layers you have set.

If a terrain has a parent, it will automatically update when the parent terrain updates, so using the save textures button on the main terrain will update all the children aswell.

**Important Details:**

- One parent terrain can be assigned to as many child terrains as you like
- Terrain Materials should only be synced to one set of terrain layers, but you can use the parent terrain field to get around that
- The parent terrain will automatically detach when you either change the terrain layers, or update the material

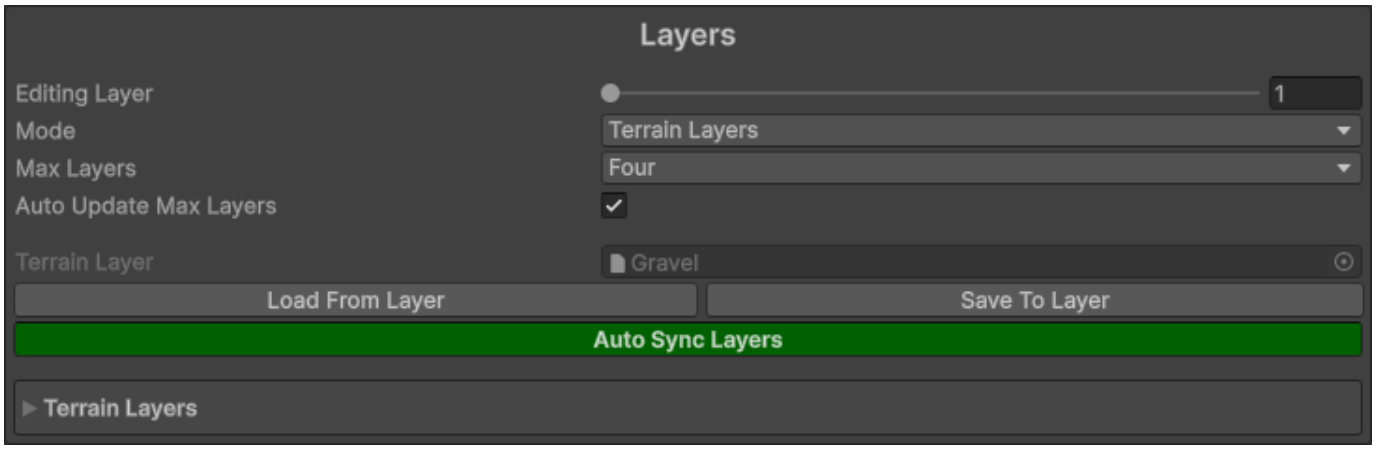

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On terrains with the RepetitionlessTerrain, when using the Create Neighbour Terrains tool and creating a directly adjacent terrain, it will automatically add a RepetitionlessTerrain component to that terrain with the selected one as the parent. This will not work for non-adjacent terrains though as shown in the image above


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## 8.2.3 Editing The Terrain Layers




| Setting | Description |
|---|---|
| Editing Layer | The layer currently being edited in the inspector |
| Mode | The mode of this material. This changes the shader depending on the option with Terrain Layers changing it to a terrain compatible shader |
| Max Layers | The max layers allowed on this material. Set lower when possible for performance improvements |
| Auto Update Max Layers | Automatically updates the above Max Layers to fit your assigned layer count |

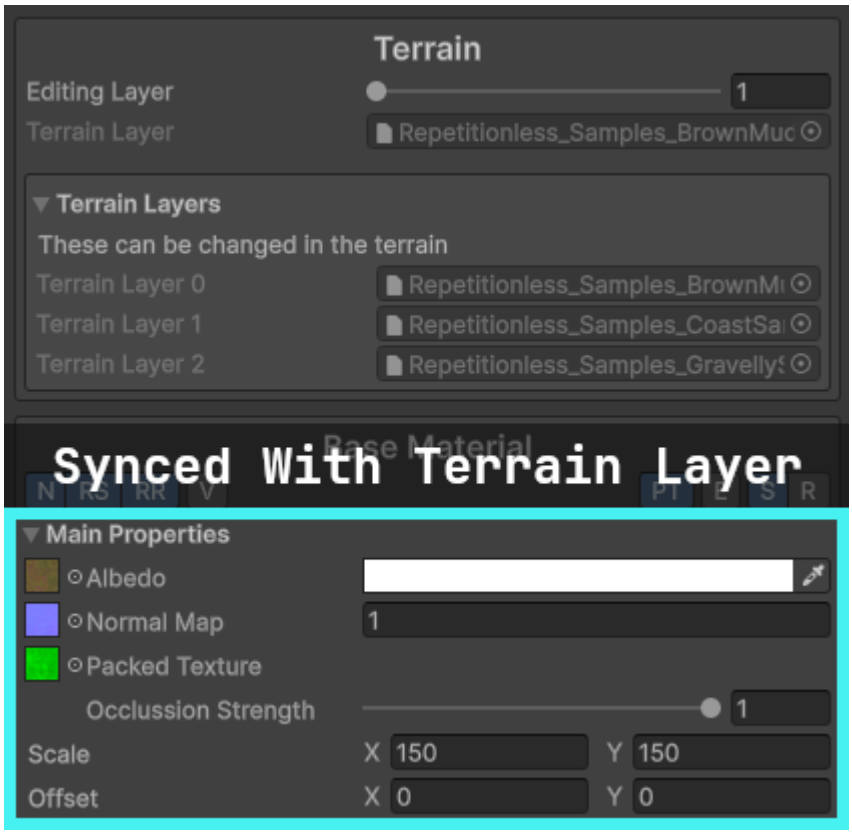
After modifying any field in the terrain layer and saving, it will automatically update those properties in any material that is linked to that layer if the auto sync is enabled. You can manually load or save to and from the layer with the buttons

The properties effected and what they link to include:

- Diffuse Map > Base Albedo Texture
- Normal Map > Base Normal Map
- Normal Scale > Base Normal Scale
- Mask Map > Base Packed Texture (Will enable packed texture when assigned)
- Metallic > Base Metallic
- Smoothness > Base Smoothness
- Tiling > Base Tiling
- Offset > Base Offset


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### 8.2.4 Editing The Material




If a terrain has been assigned, when editing the material it will show the terrain layers that it is linked to and each layer can be edited individually.

**Updating the Main Properties in the Base Material that have an equivalent in the terrain layer (listed above) will also automatically update the terrain layer**




Layers in the repetitionless material correspond to the order of the assigned terrain layers as shown in the image above

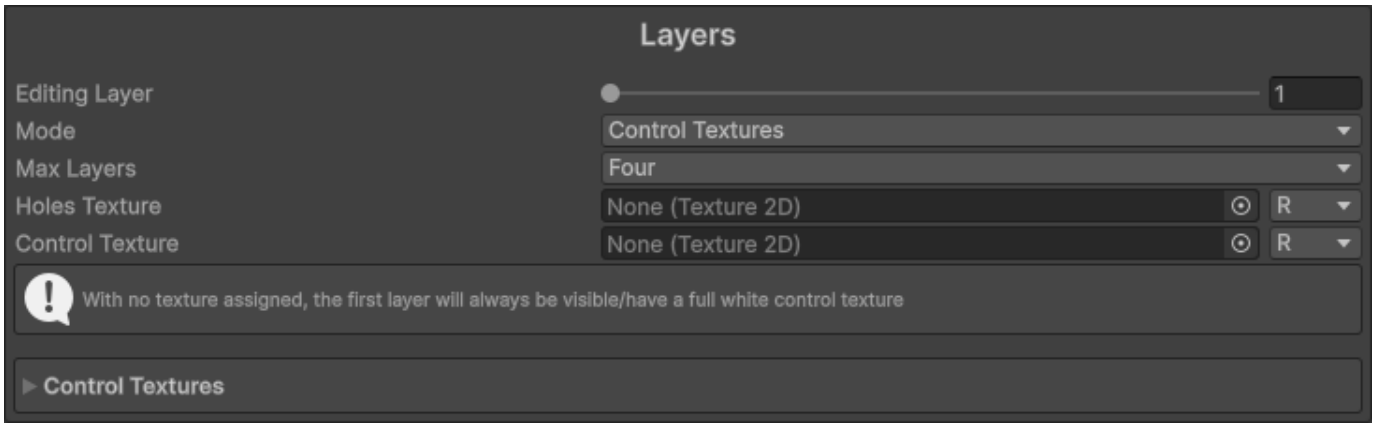
**The shader supports up to 32 terrain layers. Any more wont work and will appear white**

## 8.3 Using Control Textures

To use control textures, first make sure the layer mode is set to `Control Textures`



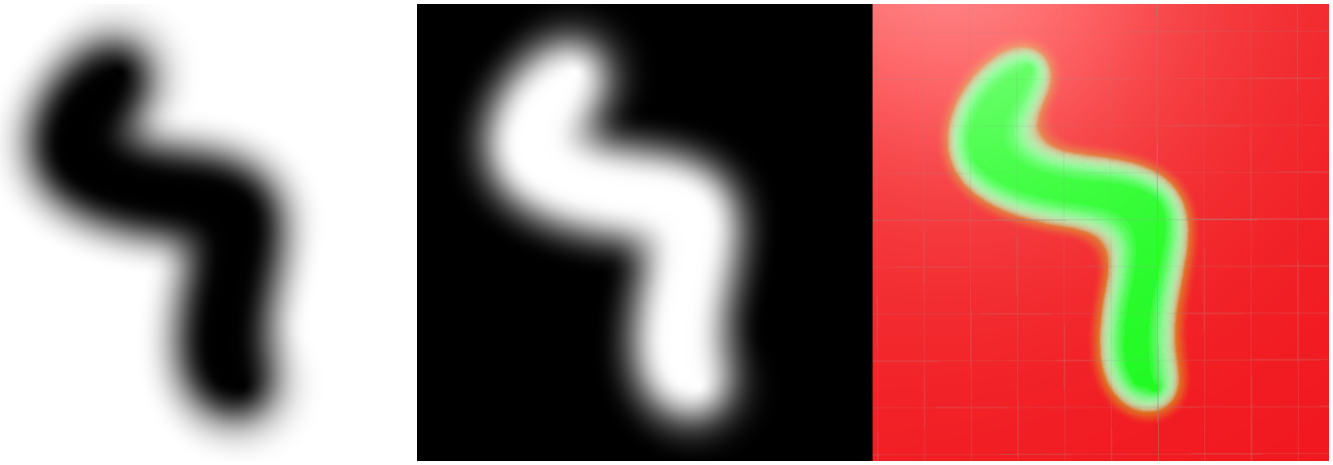
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| Setting | Description |
|---|---|
| Editing Layer | The layer currently being edited in the inspector |
| Mode | The mode of this material. This changes the shader depending on the option with Terrain Layers changing it to a terrain compatible shader |
| Max Layers | The max layers allowed on this material. Set lower when possible for performance improvements |
| Holes Texture | The holes texture that will be used for the material. It will read from the selected channel |
| Control Texture | The control texture that will be used for this layer. It will read from the selected channel |

The control textures for a specific pixel must add up to 1. For example, if you want a spot to be the second layer, the second control texture will have a value of 1 in that spot where other layers will have a value of 0.

*Example of two control textures where the left is layer 1, and the right layer 2*



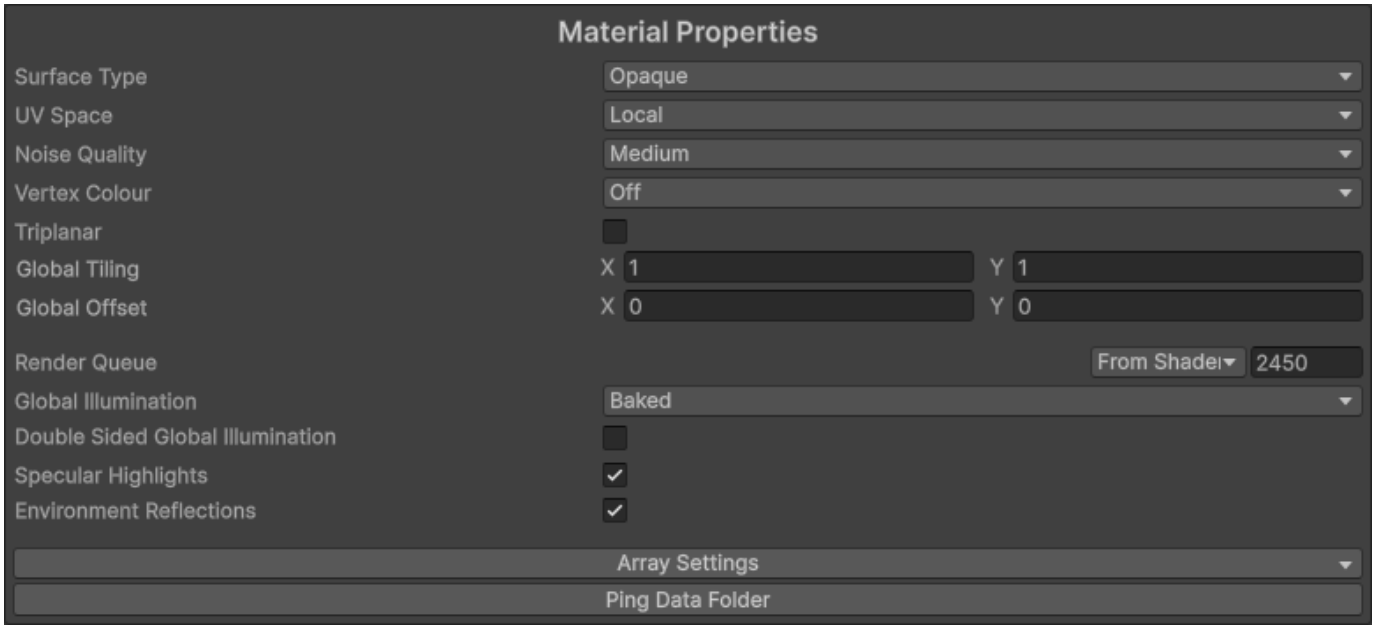
***Everything else is the same as the regular repetitionless material. To view what each property does, visit the Material Properties page***



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# 9. Material Properties

## 9.1 Material Properties






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Has all the general properties for the material

| Property | Description |
|---|---|
| Surface Type | The surface type of the material<br>0: Opaque<br>1: Cutout<br>2: Transparent |
| UV Space | Sets the UV Space<br>0: Local<br>1: World<br>*Note that with world space UVs you will require larger Tiling* |
| Noise Quality | How the noise is sampled<br>High: Dynamically samples noise<br>*(Required for non-repeating noise and the angle offset setting)*<br>Medium: Uses the pre-rendered 4k texture<br>Low: Uses the pre-rendered 1k texture |
| Vertex Colour | How the vertex colour is blended with the output colour<br>0: Off<br>1: Multiply<br>2: Additive<br>3: Subtractive<br>4: Overwrite |
| Triplanar | Enables triplanar sampling and forces world UVs<br>*Note that the first time this is set for a material it will take some time to finish as it will recompile the shader, it will be instant for subsequent toggles* |
| Global Tiling | This is multiplied with each materials tiling |
| Global Offset | This is added onto each of the materials offset |
| Render Queue | Changes when the material is drawn |
| Global Illuminattion | Controls if the global illumination is baked or realtime<br>0: Realtime<br>1: Baked<br>2: None |
| Double Sided Global Illumination | If the lightmapper accounts for both sides of the geometry when calculating Global Illumination |
| Specular Highlights | When enabled, the Material reflects the shine from direct lighting |
| Environment Reflections | When enabled, the Material samples reflections from the nearest Reflection Probes or Lighting Probe |
| Array Settings | Shows a dropdown with buttons to modify each of the texture arrays that the textures are stored in<br>AVTextures: Albedo (rgb), Variation (a)<br>NSOTextures: Normal (rg), Smoothness (b), Occlussion (a)<br>EMTextures: Emission (rgb), Metallic (a)<br>BMTextures: Blend Mask (r) |
| Ping Data Folder | Pings the data folder |

## 9.2 Toolbar



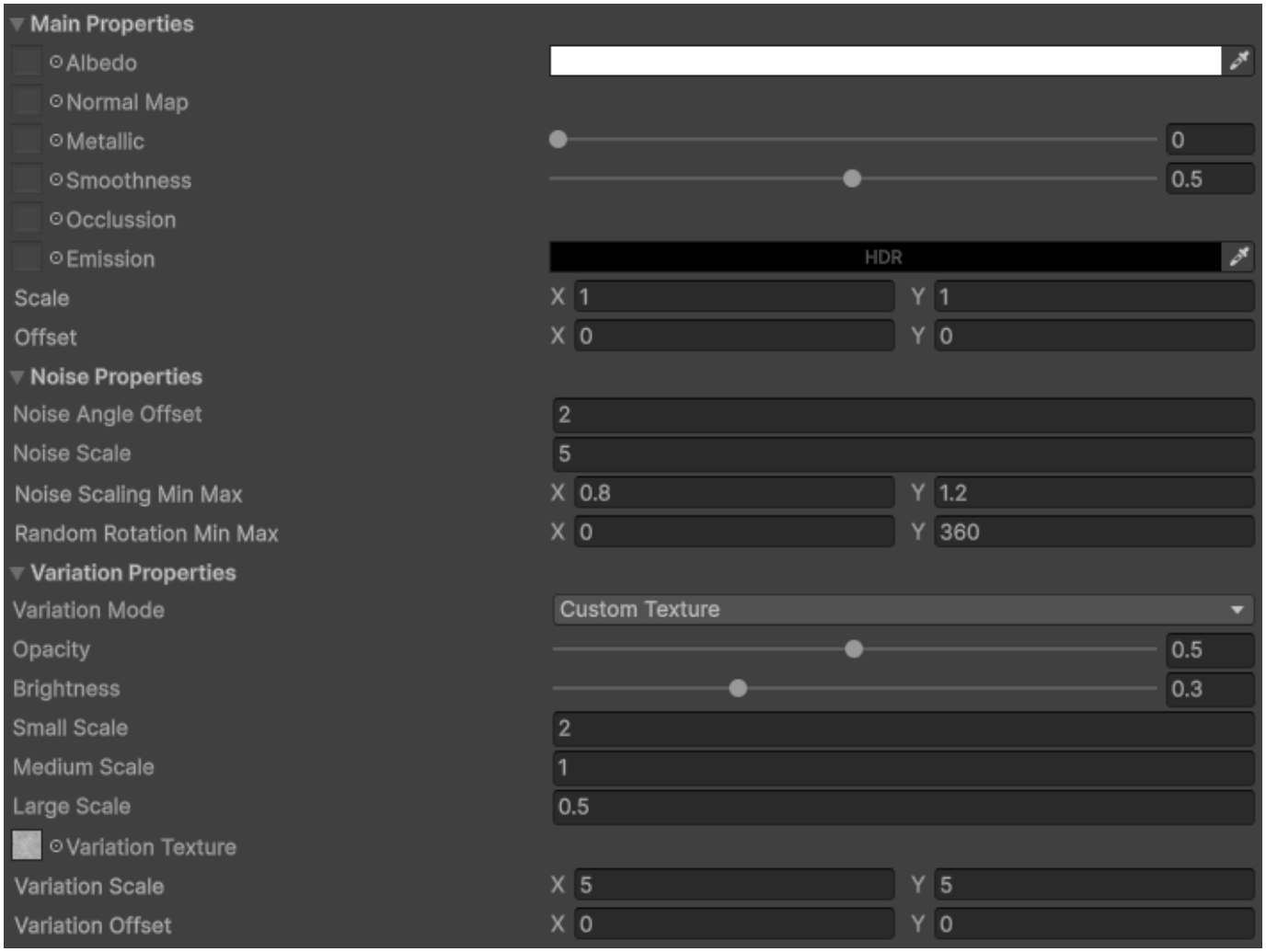

Each material has a toolbar with various settings


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Each settings text will shrink to the first letters of each word if the inspector width is too small for the whole words to fit

| Setting | Description |
|---|---|
| Noise | Adds random scaling & rotation based on voronoi noise |
| Random Scaling | Adds random scaling to each voronoi cell<br>*Only shown if Noise is enabled* |
| Random Rotation | Adds random rotation to each voronoi cell<br>*Only shown if Noise is enabled* |
| Variation | Adds random variation on top of the albedo color<br>**Using a custom texture can cause visible tiling** |
| Packed Texture | If you are using a packed texture of multiple regular ones (Better for performance)<br>R: Metallic<br>G: Occlussion<br>A: Smoothness/Roughness |
| Emission | If Emission is enabled |
| Smooth/Rough | Toggles between the smoothness and roughness texture<br>**If you are using a packed texture it will sample it with a smoothness or roughness texture based on this toggle** |

## 9.3 Material Settings






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Contains all the settings for a material. Settings can be enabled and disabled through the toolbar

This is shown for each enabled material (Base material, distance blend material, blend material)

***When a texture is assigned for fields that only use one texture channel, it will have a dropdown where you can choose which channel to use***




## 9.3.1 Main Properties

| Property | Description |
|---|---|
| Albedo | Albedo (RGB) |
| NormalMap | Normal map (RG) |
| Metallic | Metallic (R) |
| Smoothness/Roughness | Smoothness/Roughness (R)<br>*Changes between smoothness and roughness based on toolbar setting* |
| Occlussion | Occlussion (R) |
| Emission | Emission (RGB) |
| Packed Texture | The mask map for the current material<br>R: Metallic<br>G: Occlussion<br>A: Smoothness/Roughness<br>*- Only shown if Packed Texture is enabled*<br>*- Alpha toggles between smoothness and roughness based on toolbar setting* |
| Occlussion Strength | The occlussion strength of the packed texture occlussion<br>*Only shown if Packed Texture is enabled* |
| Scale | The texture tiling |
| Offset | The texture offset |

## 9.3.2 Noise Properties

These are only shown if Noise is enabled in the toolbar

| Property | Description |
|---|---|
| Noise Angle Offset | The angle offset of the voronoi noise<br>*Only shown if noise quality is set to high* |
| Noise Scale | The scale of the voronoi noise |
| Noise Scaling Min Max | Range that each voronoi cell is randomly scaled by<br>x: Min Scale<br>y: Max Scale<br>*Only shown if Random Scaling is enabled in the toolbar* |
| Random Rotation Min Max | Range that each voronoi cell is randomly rotated by<br>x: Min Rotation Degrees<br>y: Max Rotation Degrees<br>*Only shown if Random Rotation is enabled in the toolbar* |

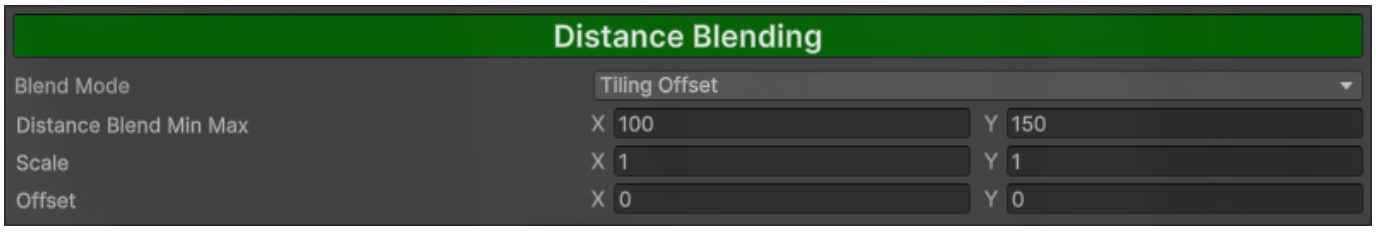

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### 9.3.3 Variation Properties

These are only shown if Variation is enabled in the toolbar

| Property | Description |
|---|---|
| Variation Mode | The variation mode used<br>Using a custom texture can cause visible tiling<br>0: Perlin noise<br>1: Simplex noise<br>2: Custom texture |
| Opacity | Transparency of the variation |
| Brightness | Intensity of the variation |
| Small Scale | Scale of the small variation sample |
| Medium Scale | Scale of the medium variation sample |
| Large Scale | Scale of the large variation sample |
| Noise Strength | Strength of the noise<br>*Only shown if Variation Mode is set to any noise* |
| Noise Scale | The noise tiling<br>*Only shown if Variation Mode is set to any noise* |
| Noise Offset | The noise offset<br>*Only shown if Variation Mode is set to any noise* |
| Variation Texture | Texture that is drawn onto other materials, can cause visible tiling<br>Variation (R), other channels are ignored<br>*Only shown if Variation Mode is set to Custom Texture* |
| Variation Scale | The variation tiling<br>*Only shown if Variation Mode is set to Custom Texture* |
| Variation Offset | The variation offset<br>*Only shown if Variation Mode is set to Custom Texture* |

## 9.4 Distance Blending



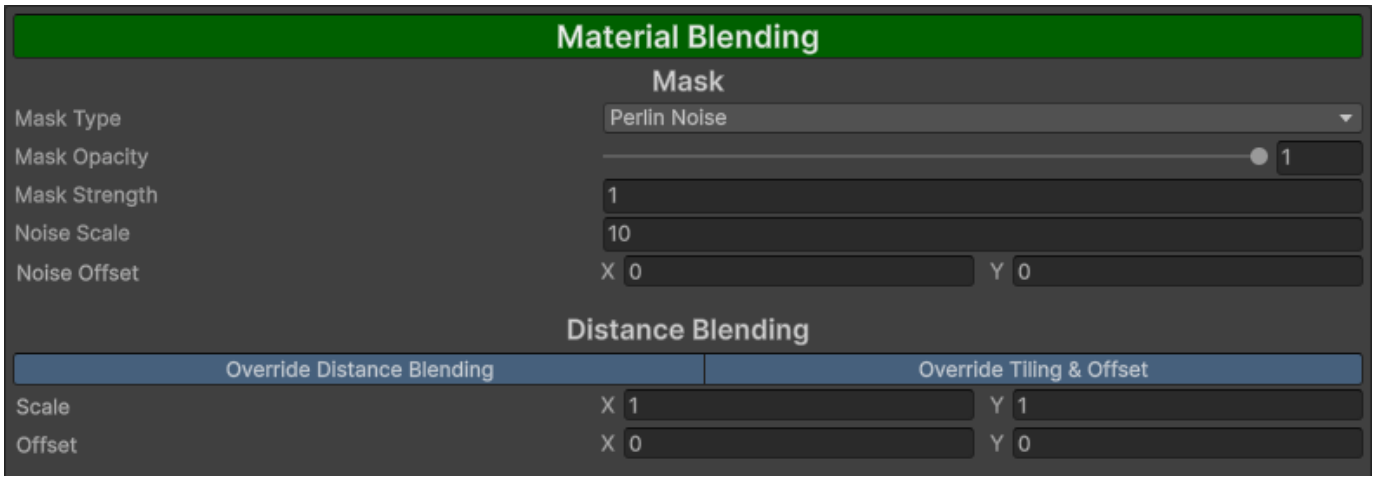

This is the material that will fade into the distance based on Distance Blend Min Max. Can be toggled by clicking the Distance Blending button


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If the Blend Mode is set to Material, a separate Material Settings will be shown below for the far material

| Property | Description |
|---|---|
| Blend Mode | Tiling Offset: Resamples materials with different Tiling and Offset<br>Material: Samples a seperate far material |
| Distance Blend Min Max | Blend distance which the material will be sampled. Materials will be blended with regular material at min, and far material at max<br>x: Min Distance<br>y: Max Distance |
| Scale | The far tiling<br>*Only shown if Blend mode is set to Tiling Offset* |
| Offset | The far offset<br>*Only shown if Blend mode is set to Tiling Offset* |

## 9.5 Material Blending




This is a separate material that will overlay the base and far material if set based on a mask. Can be toggled by clicking the Material Blending button


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## 9.5.1 Mask Settings

| Property | Description |
|---|---|
| Mask Type | The mask type used<br>0: Perlin noise<br>1: Simplex noise<br>2: Custom texture<br>3: Vertex Colour |
| Mask Opacity | Opacity of the mask and in response the blend material |
| Mask Strength | The higher the value, the sharper the edges and vice versa |
| Noise Scale | The noise tiling<br>*Only shown if Mask Type is set to any noise* |
| Noise Offset | The noise offset<br>*Only shown if Mask Type is set to any noise* |
| Blend Mask | Texture that is sampled as the mask for the blend material. Color from black-white represents opacity (0-1) Blend Mask (R), other channels are ignored<br>*Only shown if Mask Type is set to Custom Texture* |
| Scale | The blend mask tiling<br>*Only shown if Mask Type is set to Custom Texture* |
| Offset | The blend mask offset<br>*Only shown if Mask Type is set to Custom Texture* |

*If mask type is set to any noise*

| Property | Description |
|---|---|
| Noise Scale | The noise tiling |
| Noise Offset | The noise offset |

*If mask type is set to Custom Texture*

| Property | Description |
|---|---|
| Blend Mask | Texture that is sampled as the mask for the blend material. Color from black-white represents opacity (0-1) Blend Mask (R), other channels are ignored |
| Scale | The blend mask tiling |
| Offset | The blend mask offset |

*If mask type is set to Vertex Colour*

| Property | Description |
|---|---|
| Mask Colour | The colour used as a mask in the vertex colour |
| Colour Threshold | The smoothstep min max for the colour |

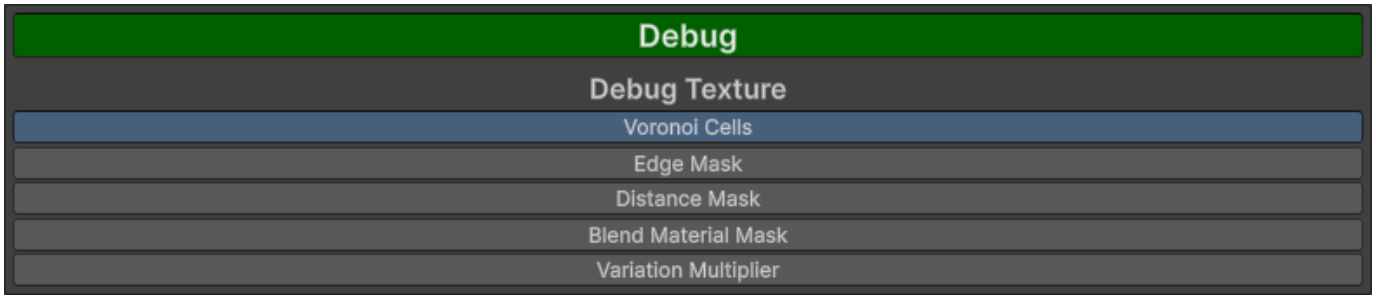

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### 9.5.2 Distance Blending Settings

These are only shown if distance blending is enabled

| Property | Description |
|---|---|
| Override Distance Blending | Draws the blend material on top of the far material |
| Override Tiling & Offset | Uses defined Tiling & Offset rather than distance blend Tiling & Offset |
| Scale | The blend mask distance scale<br>*Only shown if Override Tiling & Offset is enabled* |
| Offset | The blend mask distance offset<br>*Only shown if Override Tiling & Offset is enabled* |

## 9.6 Debug




This will show the values of the selected option. Can be toggled by clicking the Debug button

| Option | Description |
|---|---|
| Voronoi Cells | Outputs the value of each voronoi cell |
| Edge Mask | Outputs the edge mask for the voronoi cells |
| Distance Mask | Outputs the distance mask |
| Blend Material Mask | Outputs the blend material mask |
| Variation Multiplier | Outputs the variation multiplier |


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# 10. Painting

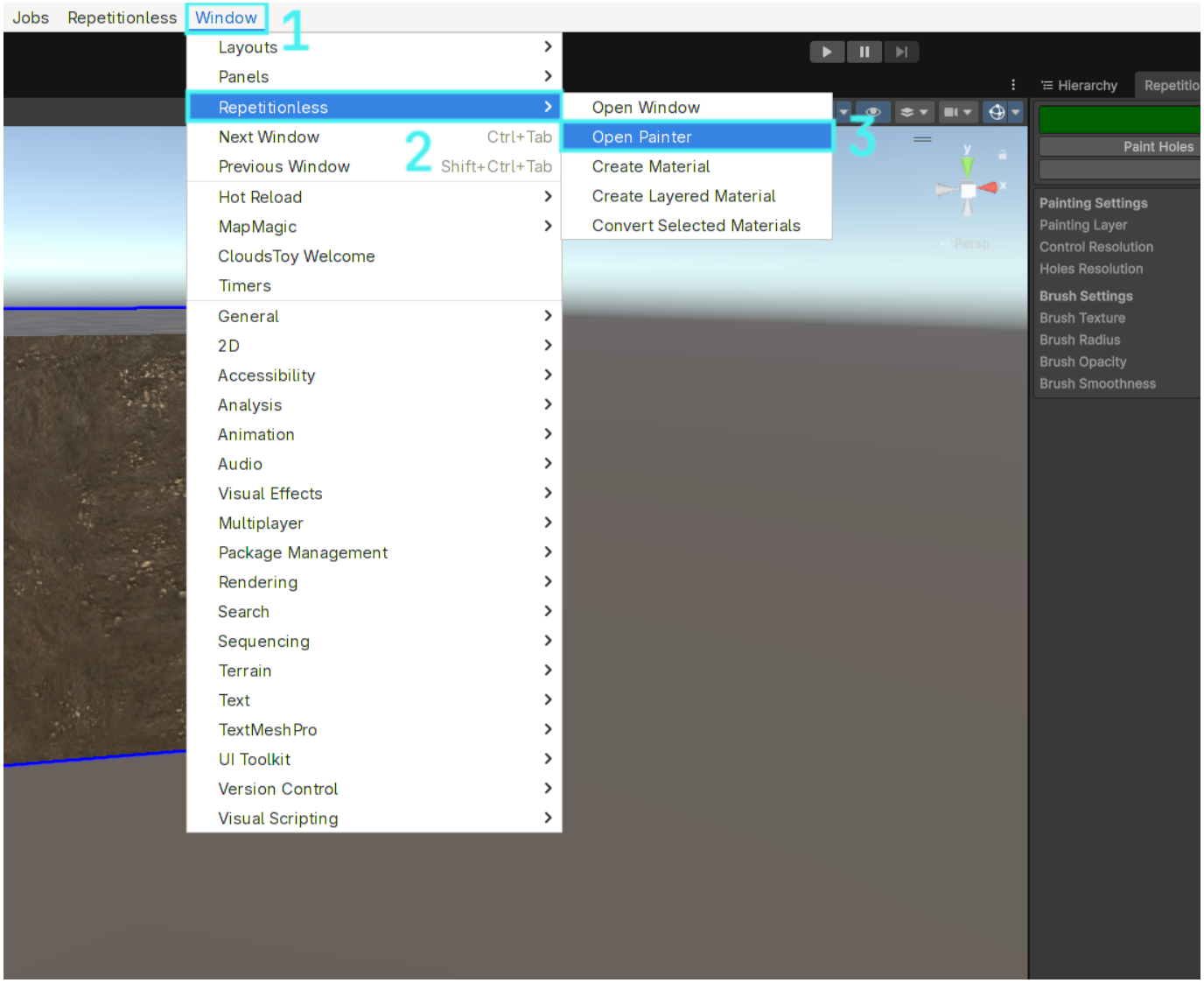
## 10.1 Tutorial Video

https://youtu.be/klj2tto107A

## 10.2 Setting Up

You can open the painter window in the toolbar:

`Window > Repetitionless > Open Painter`




Painting is only supported on objects that have the following:

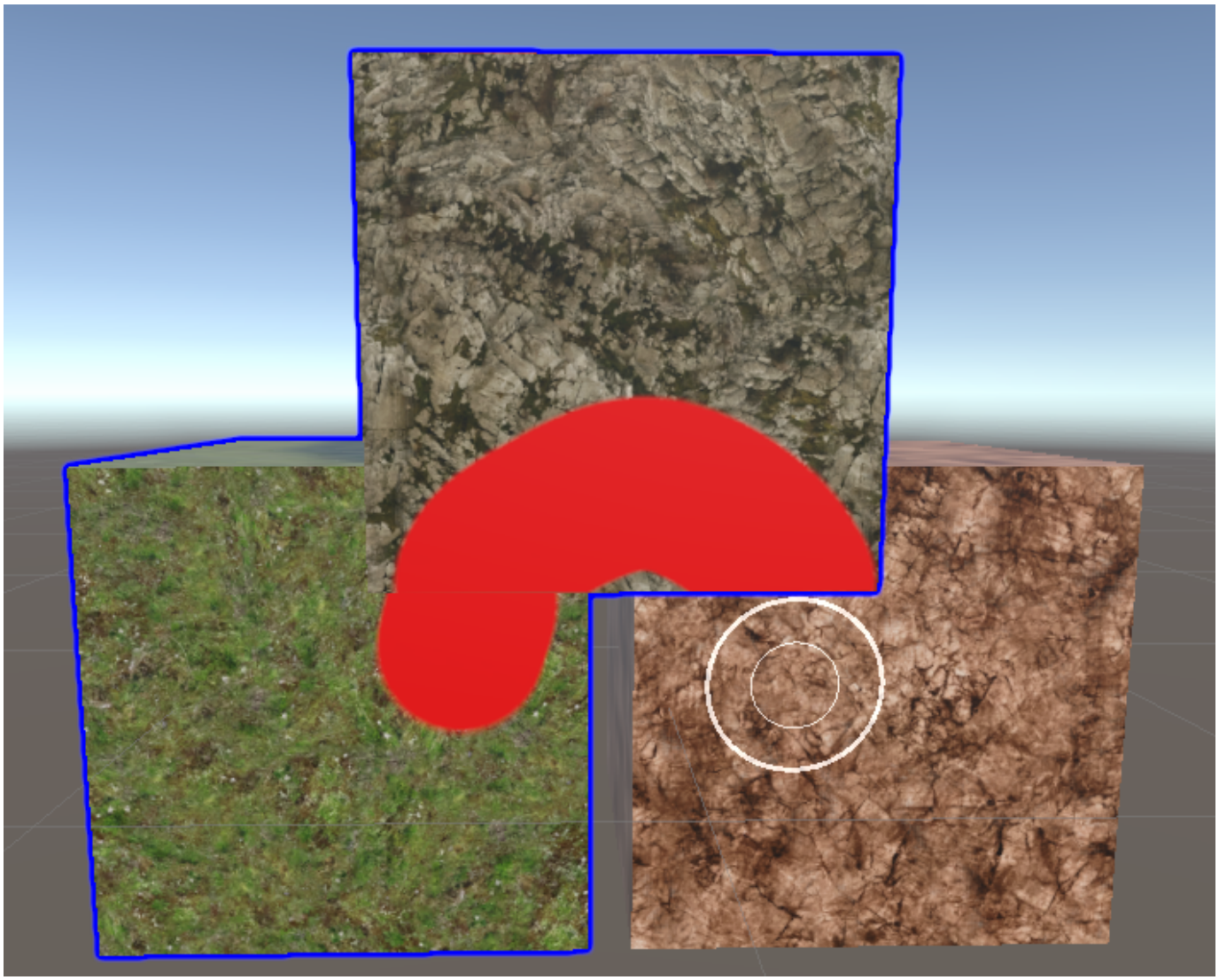
- A mesh renderer with a Repetitionless Layered material (If the layer mode is not set to control textures it will automatically be converted)
- A mesh collider

Trying to paint any other object will do nothing



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## 10.3 Selection



When painting is enabled, clicking an object will automatically select it and start painting if the object can be painted

Objects that are selected for painting will have a dark blue outline around them. Trying to paint outside of any selected objects in one stroke will do nothing even if the unselected hovered object can be painted. While painting, any unselected objects will be passed through and selected objects can still be painted behind them. It basically works as a mask

Regular object selection while painting is partially disabled. When selecting objects in the scene view it will use the painter selection, but you can select objects normally in the Hierarchy window

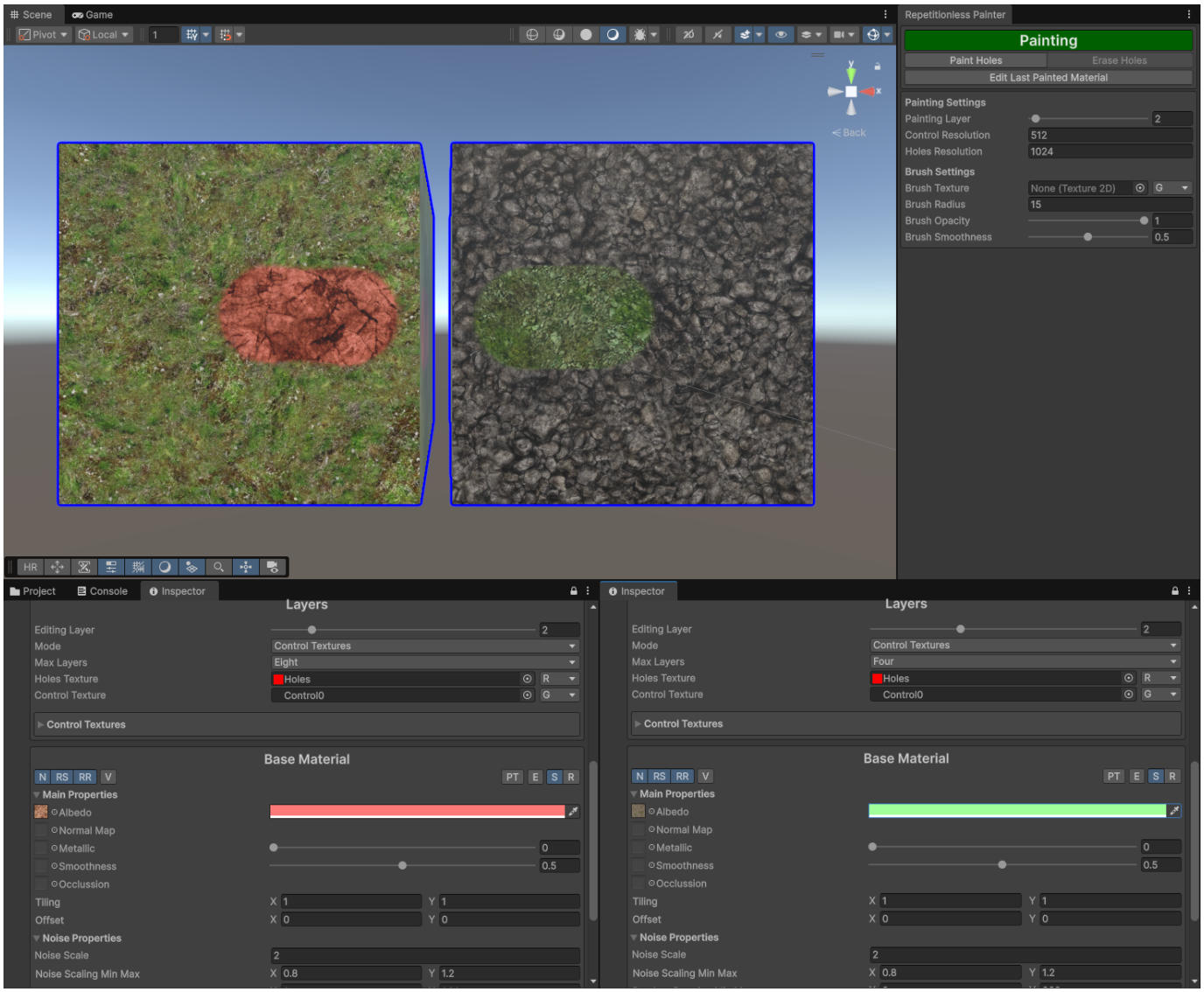
When an object is first selected, all the control textures will be created/resized, and the holes texture will be created/resized. This can take some time depending on your control/holes resolution settings

## 10.4 Layers

The layer selection in the window is the selected layer for each material. This is per material and can be edited in its inspector. For example, layer 1 on one object could be different to layer 1 on another object as shown below



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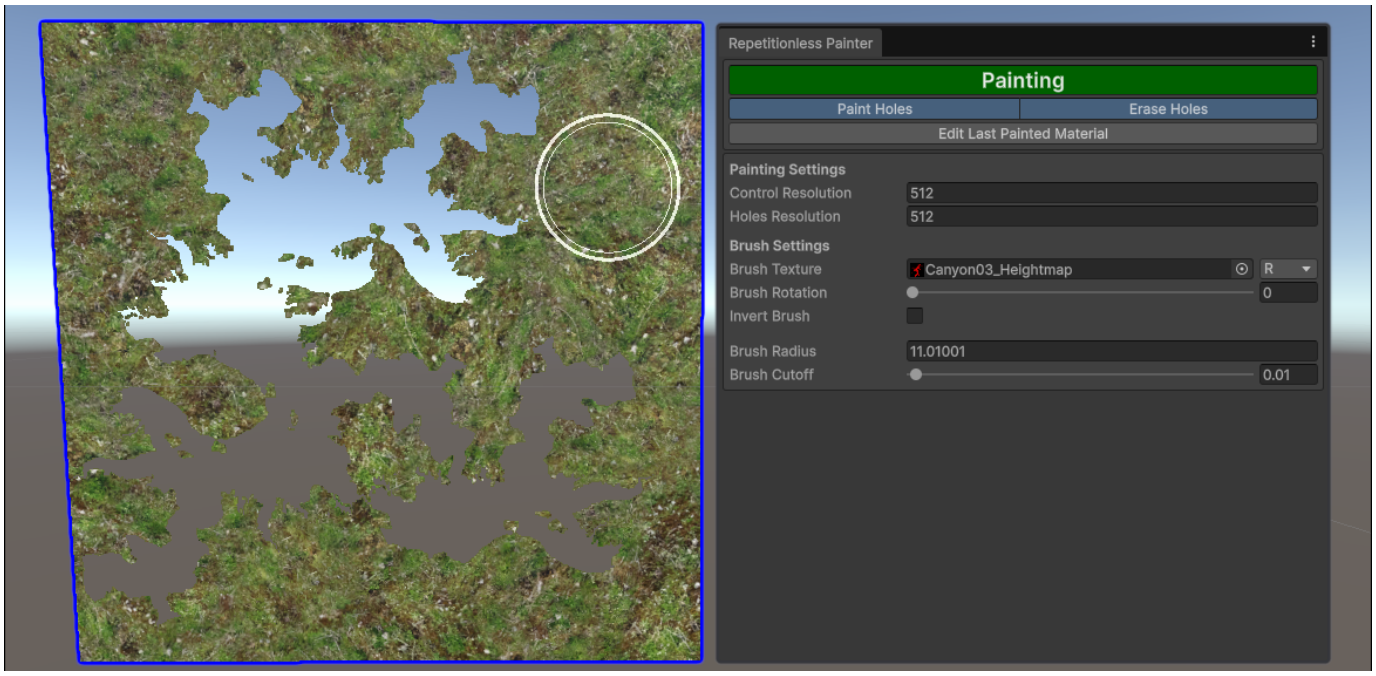

## 10.5 Holes

Holes can be painted onto any object with the Paint Holes toggle

When Erase holes is enabled, painting over existing holes will erase them. This is visualised with an inner circle inside the brush when enabled

Since holes do not have transparency, when using a brush texture, you can adjust the cutoff to determine which parts of the texture will be counted as a hole. If a value at a given pixel is less than the cutoff, it will not be counted as a hole

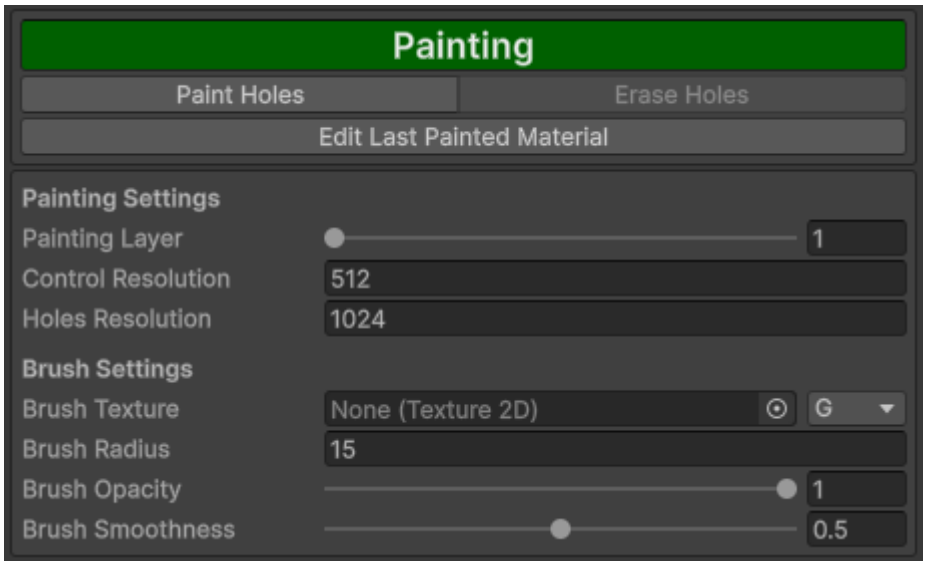

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## 10.6 Properties

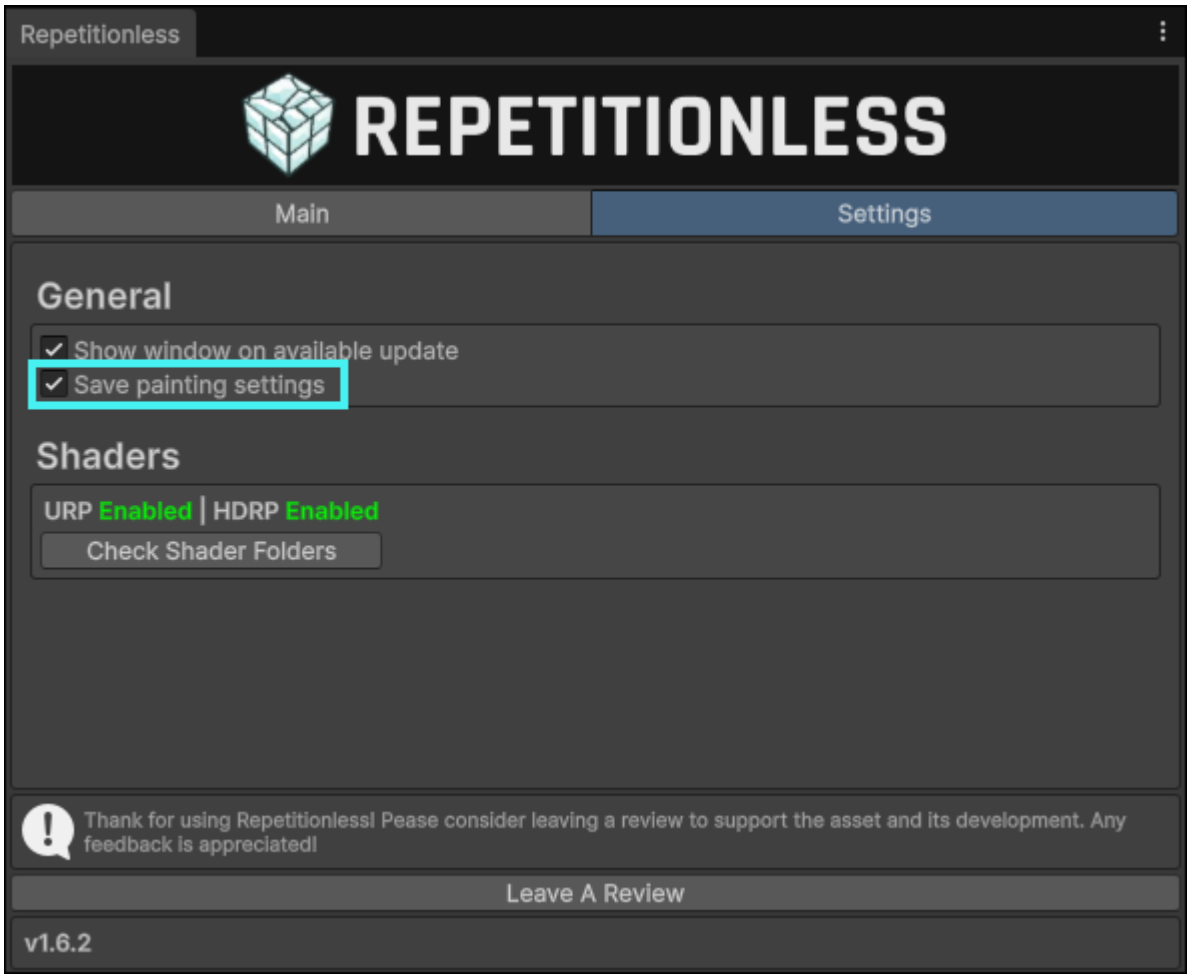



| Property | Description |
|---|---|
| Painting Layer | The layer that will be painted. This is determined per material in its layer selection |
| Control Resolution | **Default:** 512<br>The resolution of the control textures. Existing textures will be automatically resized to this resolution |
| Holes Resolution | **Default:** 512<br>The resolution of the holes texture. Existing textures will be automatically resized to this resolution |
| Brush Texture | The texture used for painting, if not set it will be a circle filling the radius.<br>The channel is what channel to read from in the texture |
| Brush Rotation | The rotation of the brush texture relative to the uvs of the painted object. This does nothing with no texture set |
| Invert Brush | Inverts the brush texture, sampling it as (1 - value) |
| Brush Radius | The size of the brush |
| Brush Cutoff | The cutoff of the value for the brush texture where holes will not be drawn |
| Brush Opacity | he strength of the brush. The brush will accumulate so if you want opacity while dragging, set this to <= 0.05 |
| Brush Smoothness | What radius to start fading out the brush. This is visualised as the inner circle in the scene view |

All properties can be auto saved and kept between painting sessions (enabled by default). You can toggle this by:

1. Opening the main window in the toolbar at `Window > Repetitionless > Open`
2. Click the settings tab
3. Toggle the setting called `Save painting settings`


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## 10.7 Keybinds




When using keybinds, they will always show with a popup next to the mouse so you know what you have just changed without looking at the window. This allows painting objects without having the window visible at all


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Note that:

- Keybinds will only work when the scene view is focused
- Default actions with the below keybinds will be disabled while painting.

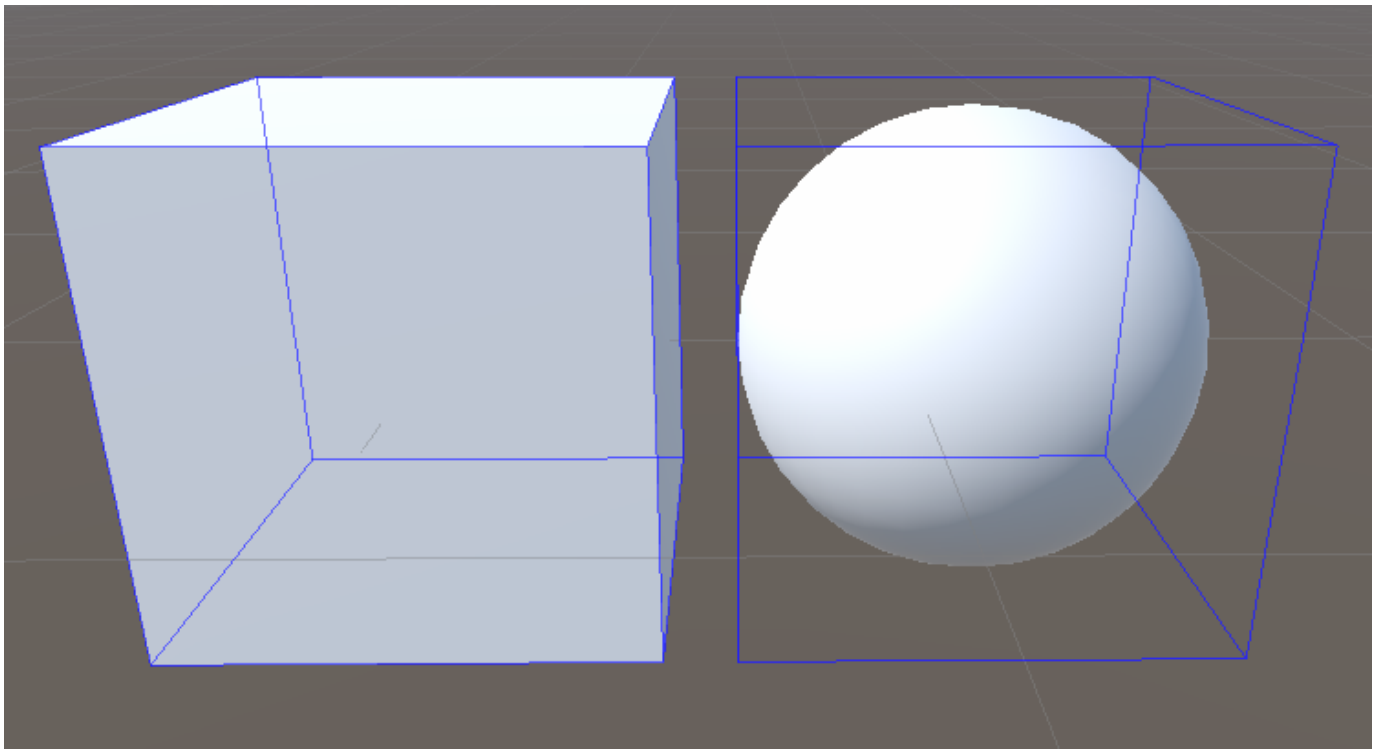
| Key | Action |
|---|---|
| A | **Painting Control:** Changes the brush opacity<br>**Painting Holes:** Changes the brush cutoff |
| S | Changes the brush Radius |
| D | **Painting Control:** Changes the brush smoothness<br>**Painting Holes**: Toggles erasing holes |
| C | Changes the brush texture rotation |
| F | Focuses the scene camera to the mouse position |
| G | Toggles painting |
| H | Toggles painting holes |
| Shift + Resize | **Compatible with Opacity, Cutoff, Radius, Smoothness, & Rotation controls**<br>Slows down the modification of a slideable property to x0.1 the default. Also shows an extra digit in the mouse popup |
| Shift + Click | **Hovered Selected:** Deselects the hovered object<br>**Hovered Unselected:** Selects the hovered object without painting |
| Shift + Scroll | Changes the painting layer |
| Control (Hold) | Inverts the brush |

## 10.8 Caveats

### 10.8.1 Pre Unity 2022

Below Unity 2022 there is no cheap built in way to draw outlines around the selected objects. In these versions a wire cube is drawn around the bounds of selected objects instead.


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## 10.8.2 UVs

A mesh must have no overlapping or mirrored UVs for painting to properly work everywhere. If there is any overlapping or mirrored UVs, painting will only work on one of the islands, and for any other islands nothing will be painted.

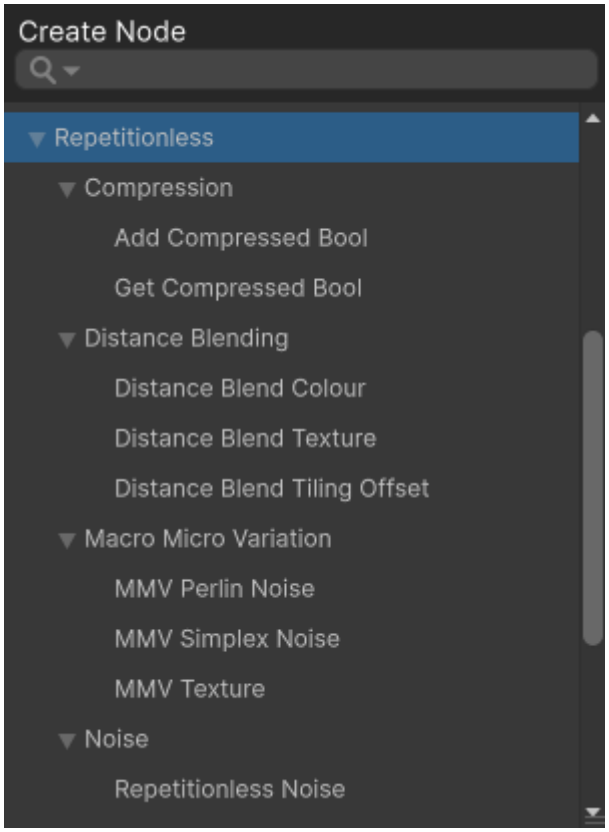
Note that: - Painting on the island that works will reflect to all other overlapping islands - The island that is paintable is unknown as it is the last one that was baked



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# 11. Shader Creation

## 11.1 Shader Graphs




To create a shader graph using the included sub-shaders, you can find the sub shaders in the shader graph under `Create Node > Repetitionless`

To view details about each sub-shader, view the corresponding Sub Graphs page in the contents on the left of the page

## 11.2 Shader Code

For creating your own shaders using the included shaders, view the respective Shader API pages in the contents on the left of the page for details on each set of functions and how to implement them into your own code

Shaders can be found at `Packages/com.williamschack.repetitionless/Shaders/HLSL`



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# 12. Scripting

For creating your own scripts and inspectors using the included scripts, view the respective Scripting API pages in the contents on the left of the page for details on each set of functions and how to implement them into your own code

Scripts can be found at `Packages/com.williamschack.repetitionless` in the `Editor` and `Runtime` folders

**To include script namespaces you must reference their respective assembly definitions in your own assembly definitions**


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# 13. Submitting Issues

If you would like to submit a Bug Report, Feature Request, or Question, you can:

- Submit an issue to the discord (https://discord.gg/2NTvBDB3xq))
- Submit an issue to the github repo (github.com/WilliamSchack/Repetitionless-Issues)
- Email me at (support@wilschack.dev)


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# 14. Integrations

## 14.1 MapMagic

`MapMagic2 Must Be Installed`

### 14.1.1 Applying Repetitionless

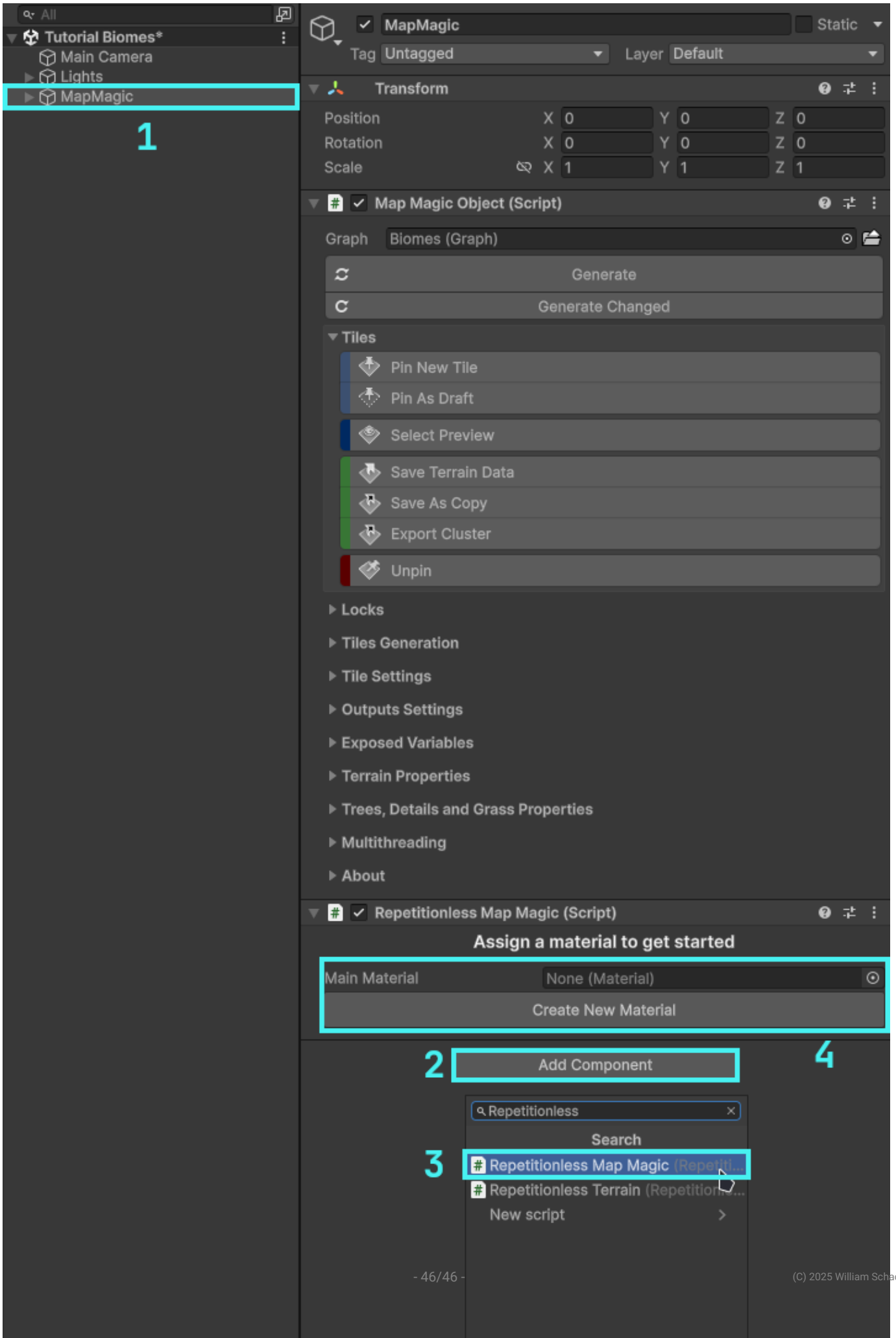
To use repetitionless on a MapMagic terrain, add a `RepetitionlessMapMagic` component to the object that has the MapMagicObject

1. Select the map magic terrain you want to use
2. Select `Add Component`
3. Select `Repetitionless Map Magic`

To add a material you can either:

- Click the `Create New Material` button
- Assign a material in the `Main Material` field


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